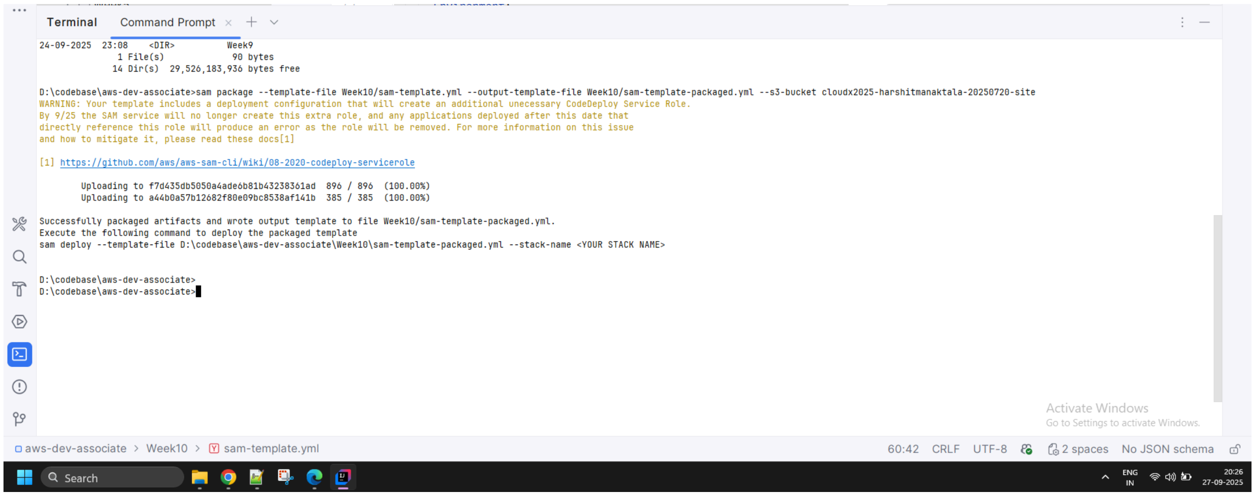
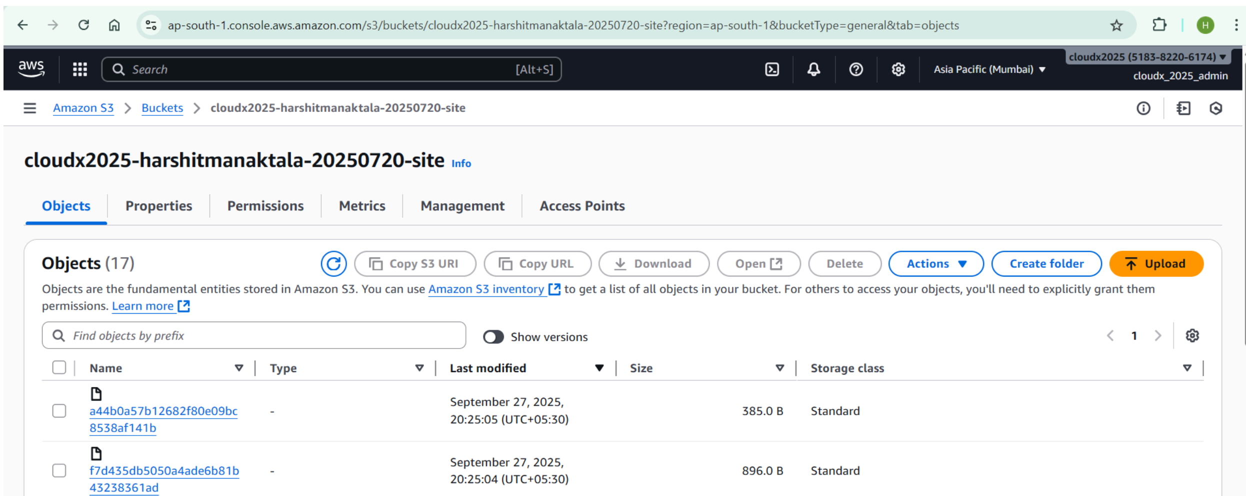
Sub-task 1 – Create and deploy SAM application

1. Create a SAM application for the *<ProjectName>-UploadsNotificationFunction* from the Module 10: Serverless:
    - Create a SAM template for the lambda function and store it in the root of your project
    - Specify IAM roles, required for integration with SNS, SQS from Module 9: SNS/SQS
    - Specify IAM role to allow storing logs in CloudWatch
    - Following this tutorial, define any *DeploymentPreferenceType* you wish
    - *Optionally*: specify in the template additional Lambda functions for testing the SAM (you can choose *<ProjectName>-S3LogsFunction* from the Module 10: Serverless) deployment
2. Delete the *<ProjectName>-UploadsNotificationFunction* from Module 10: Serverless.
3. Following this tutorial, install SAM CLI on your local machine.
4. Using the terminal on your local machine, execute sam deploy CLI command with appropriate flags to store the packaged lambda to S3 and deploy it with the appropriate resources. Under the packaged lambda is considered some executable artifact(for example jar file for Java).
5. In AWS console, open Lambda service and find in Functions section your deployed lambda function. Also, in Applications tab, find your deployed SAM application and review all created resources.
6. Upload a new image to the image S3 bucket using your web-application and ensure an email notification which mentions the image is sent.

Command#1: `sam package --template-file Week10/sam-template.yml --output-template-file Week10/sam-template-packaged.yml --s3-bucket cloudx2025-harshitmanaktala-20250720-site`





Command#2: `sam deploy --template-file Week10/sam-template-packaged.yml --stack-name cloudx2025-lambdastack --capabilities CAPABILITY_IAM CAPABILITY_NAMED_IAM --parameter-overrides ProjectName=cloudx2025 S3BucketName=cloudx2025-ab317d87-5281-4b42-be36-431c0fc080aa SnsTopicArn=arn:aws:sns:ap-south-1:518382206174:cloudx2025-UploadsNotificationTopic SqsQueueArn=arn:aws:sqs:ap-south-1:518382206174:cloudx2025-UploadsNotificationQueue SqsQueueUrl=https://sqs.ap-south-1.amazonaws.com/518382206174/cloudx2025-UploadsNotificationQueue`

Terminal   Command Prompt

```
D:\codebase\aws-dev-associate>sam deploy --template-file Week10/sam-template-packaged.yml --stack-name cloudx2025-lambdastack --capabilities CAPABILITY_IAM CAPABILITY_NAMED_IAM --parameter-overrides ProjectName=cloudx2025 S3BucketName=cloudx2025-ab317d87-5281-4b42-be36-431c0fc080aa SnsTopicArn=arn:aws:sns:ap-south-1:518382206174:cloudx2025-UploadsNotificationTopic SqsQueueArn=arn:aws:sqs:ap-south-1:518382206174:cloudx2025-UploadsNotificationQueue SqsQueueUrl=https://sqs.ap-south-1.amazonaws.com/518382206174/cloudx2025-UploadsNotificationQueue

        Deploying with following values
        ===============================
        Stack name                   : cloudx2025-lambdastack
        Region                       : None
        Confirm changeset            : False
        Disable rollback             : False
        Deployment s3 bucket         : None
        Capabilities                 : ["CAPABILITY_IAM", "CAPABILITY_NAMED_IAM"]
        Parameter overrides          : {"ProjectName": "cloudx2025", "S3BucketName": "cloudx2025-ab317d87-5281-4b42-be36-431c0fc080aa", "SnsTopicArn": "arn:aws:sns:ap-south-1:518382206174:cloudx2025-UploadsNotificationTopic", "SqsQueueArn": "arn:aws:sqs:ap-south-1:518382206174:cloudx2025-UploadsNotificationQueue", "SqsQueueUrl": "https://sqs.ap-south-1.amazonaws.com/518382206174/cloudx2025-UploadsNotificationQueue"}
        Signing Profiles             : {}

Initiating deployment
=====================


Waiting for changeset to be created..

CloudFormation stack changeset
-------------------------------------------------------------------------------------------------------------------------
Operation                   LogicalResourceId                               ResourceType                        Replacement
-------------------------------------------------------------------------------------------------------------------------
+ Add                       CodeDeployServiceRole                           AWS::IAM::Role                      N/A
+ Add                       S3LogsFunctionRole                              AWS::IAM::Role                      N/A
+ Add                       S3LogsFunction                                  AWS::Lambda::Function               N/A
+ Add                       ServerlessDeploymentApplication                 AWS::CodeDeploy::Application        N/A
+ Add                       UploadsNotificationFunctionAliaslive            AWS::Lambda::Alias                  N/A
+ Add                       UploadsNotificationFunctionDeploymentGroup      AWS::CodeDeploy::DeploymentGroup    N/A
+ Add                       UploadsNotificationFunctionSQSTrigger           AWS::Lambda::EventSourceMapping     N/A
+ Add                       UploadsNotificationFunctionVersion50cdb2704d    AWS::Lambda::Version                N/A
```

Activate Windows
Go to Settings to activate Windows.

aws-dev-associate > Week10 > sam-template-packaged.yml   116:36 CRLF UTF-8 2 spaces No JSON schema

Search   ENG IN   22:10 27-09-2025

Terminal   Command Prompt

```
+ Add                       S3LogsFunction                                  AWS::Lambda::Function               N/A
+ Add                       ServerlessDeploymentApplication                 AWS::CodeDeploy::Application        N/A
+ Add                       UploadsNotificationFunctionAliaslive            AWS::Lambda::Alias                  N/A
+ Add                       UploadsNotificationFunctionDeploymentGroup      AWS::CodeDeploy::DeploymentGroup    N/A
+ Add                       UploadsNotificationFunctionSQSTrigger           AWS::Lambda::EventSourceMapping     N/A
+ Add                       UploadsNotificationFunctionVersion50cdb2704d    AWS::Lambda::Version                N/A
+ Add                       UploadsNotificationFunction                     AWS::Lambda::Function               N/A
+ Add                       UploadsNotificationLambdaRole                   AWS::IAM::Role                      N/A
-------------------------------------------------------------------------------------------------------------------------


Changeset created successfully. arn:aws:cloudformation:ap-south-1:518382206174:changeSet/samcli-deploy1758991151/36869470-4423-4416-a736-9236328c0692


2025-09-27 22:09:17 - Waiting for stack create/update to complete

CloudFormation events from stack operations (refresh every 5.0 seconds)
-------------------------------------------------------------------------------------------------------------------------
ResourceStatus              ResourceType                        LogicalResourceId                               ResourceStatusReason
-------------------------------------------------------------------------------------------------------------------------
CREATE_IN_PROGRESS          AWS::CloudFormation::Stack          cloudx2025-lambdastack                          User Initiated
CREATE_IN_PROGRESS          AWS::IAM::Role                      CodeDeployServiceRole                           -
CREATE_IN_PROGRESS          AWS::CodeDeploy::Application        ServerlessDeploymentApplication                 -
CREATE_IN_PROGRESS          AWS::IAM::Role                      S3LogsFunctionRole                              -
CREATE_IN_PROGRESS          AWS::IAM::Role                      UploadsNotificationLambdaRole                   -
CREATE_IN_PROGRESS          AWS::CodeDeploy::Application        ServerlessDeploymentApplication                 Resource creation Initiated
CREATE_COMPLETE             AWS::CodeDeploy::Application        ServerlessDeploymentApplication                 -
CREATE_IN_PROGRESS          AWS::IAM::Role                      CodeDeployServiceRole                           Resource creation Initiated
CREATE_IN_PROGRESS          AWS::IAM::Role                      S3LogsFunctionRole                              Resource creation Initiated
CREATE_IN_PROGRESS          AWS::IAM::Role                      UploadsNotificationLambdaRole                   Resource creation Initiated
CREATE_COMPLETE             AWS::IAM::Role                      CodeDeployServiceRole                           -
CREATE_COMPLETE             AWS::IAM::Role                      S3LogsFunctionRole                              -
CREATE_IN_PROGRESS          AWS::CodeDeploy::DeploymentGroup    UploadsNotificationFunctionDeploymentGroup      -
CREATE_IN_PROGRESS          AWS::CodeDeploy::DeploymentGroup    UploadsNotificationFunctionDeploymentGroup      Resource creation Initiated
CREATE_COMPLETE             AWS::CodeDeploy::DeploymentGroup    UploadsNotificationFunctionDeploymentGroup      -
CREATE_IN_PROGRESS          AWS::Lambda::Function               S3LogsFunction                                  -
```

Activate Windows
Go to Settings to activate Windows.

aws-dev-associate > Week10 > sam-template-packaged.yml   116:36 CRLF UTF-8 2 spaces No JSON schema

Search   ENG IN   22:10 27-09-2025

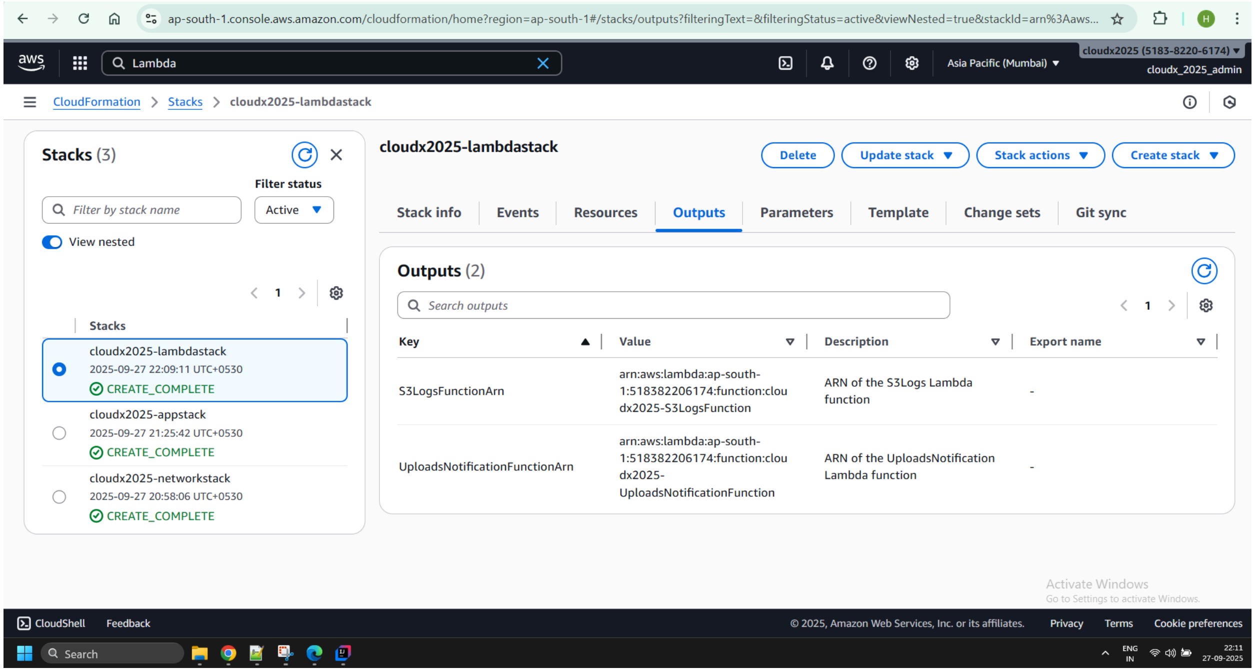
Terminal   Command Prompt

```
CREATE_IN_PROGRESS          AWS::Lambda::Function               S3LogsFunction                                  Resource creation Initiated
CREATE_COMPLETE             AWS::IAM::Role                      UploadsNotificationLambdaRole                   -
CREATE_IN_PROGRESS          AWS::Lambda::Function               UploadsNotificationFunction                     -
CREATE_IN_PROGRESS          AWS::Lambda::Function               UploadsNotificationFunction                     Resource creation Initiated
CREATE_COMPLETE             AWS::Lambda::Function               S3LogsFunction                                  -
CREATE_COMPLETE             AWS::Lambda::Function               UploadsNotificationFunction                     -
CREATE_IN_PROGRESS          AWS::Lambda::Version                UploadsNotificationFunctionVersion50cdb2704d    -
CREATE_IN_PROGRESS          AWS::Lambda::Version                UploadsNotificationFunctionVersion50cdb2704d    Resource creation Initiated
CREATE_COMPLETE             AWS::Lambda::Version                UploadsNotificationFunctionVersion50cdb2704d    -
CREATE_IN_PROGRESS          AWS::Lambda::Alias                  UploadsNotificationFunctionAliaslive            -
CREATE_IN_PROGRESS          AWS::Lambda::Alias                  UploadsNotificationFunctionAliaslive            Resource creation Initiated
CREATE_COMPLETE             AWS::Lambda::Alias                  UploadsNotificationFunctionAliaslive            -
CREATE_IN_PROGRESS          AWS::Lambda::EventSourceMapping     UploadsNotificationFunctionSQSTrigger           -
CREATE_IN_PROGRESS          AWS::Lambda::EventSourceMapping     UploadsNotificationFunctionSQSTrigger           Resource creation Initiated
CREATE_COMPLETE             AWS::Lambda::EventSourceMapping     UploadsNotificationFunctionSQSTrigger           -
CREATE_COMPLETE             AWS::CloudFormation::Stack          cloudx2025-lambdastack                          -
-------------------------------------------------------------------------------------------------------------------------

CloudFormation outputs from deployed stack
-------------------------------------------------------------------------------------------------------------------------
Outputs
-------------------------------------------------------------------------------------------------------------------------
Key                 UploadsNotificationFunctionArn
Description         ARN of the UploadsNotification Lambda function
Value               arn:aws:lambda:ap-south-1:518382206174:function:cloudx2025-UploadsNotificationFunction

Key                 S3LogsFunctionArn
Description         ARN of the S3Logs Lambda function
Value               arn:aws:lambda:ap-south-1:518382206174:function:cloudx2025-S3LogsFunction
-------------------------------------------------------------------------------------------------------------------------


Successfully created/updated stack - cloudx2025-lambdastack in None


D:\codebase\aws-dev-associate>
```

Activate Windows
Go to Settings to activate Windows.

aws-dev-associate > Week10 > sam-template-packaged.yml   116:36 CRLF UTF-8 2 spaces No JSON schema

Search   ENG IN   22:10 27-09-2025


ap-south-1.console.aws.amazon.com/cloudformation/home?region=ap-south-1#/stacks/outputs?filteringText=&filteringStatus=active&viewNested=true&stackId=arn%3Aaws...
Lambda
Asia Pacific (Mumbai)
cloudx2025 (5183-8220-6174)
cloudx_2025_admin
CloudFormation > Stacks > cloudx2025-lambdastack
Stacks (3)
Filter status
Filter by stack name
Active
View nested
Stacks
cloudx2025-lambdastack
2025-09-27 22:09:11 UTC+0530
CREATE_COMPLETE
cloudx2025-appstack
2025-09-27 21:25:42 UTC+0530
CREATE_COMPLETE
cloudx2025-networkstack
2025-09-27 20:58:06 UTC+0530
CREATE_COMPLETE
cloudx2025-lambdastack
Delete
Update stack
Stack actions
Create stack
Stack info
Events
Resources
Outputs
Parameters
Template
Change sets
Git sync
Outputs (2)
Search outputs
Key
Value
Description
Export name
S3LogsFunctionArn
arn:aws:lambda:ap-south-1:518382206174:function:cloudx2025-S3LogsFunction
ARN of the S3Logs Lambda function
-
UploadsNotificationFunctionArn
arn:aws:lambda:ap-south-1:518382206174:function:cloudx2025-UploadsNotificationFunction
ARN of the UploadsNotification Lambda function
-
CloudShell
Feedback
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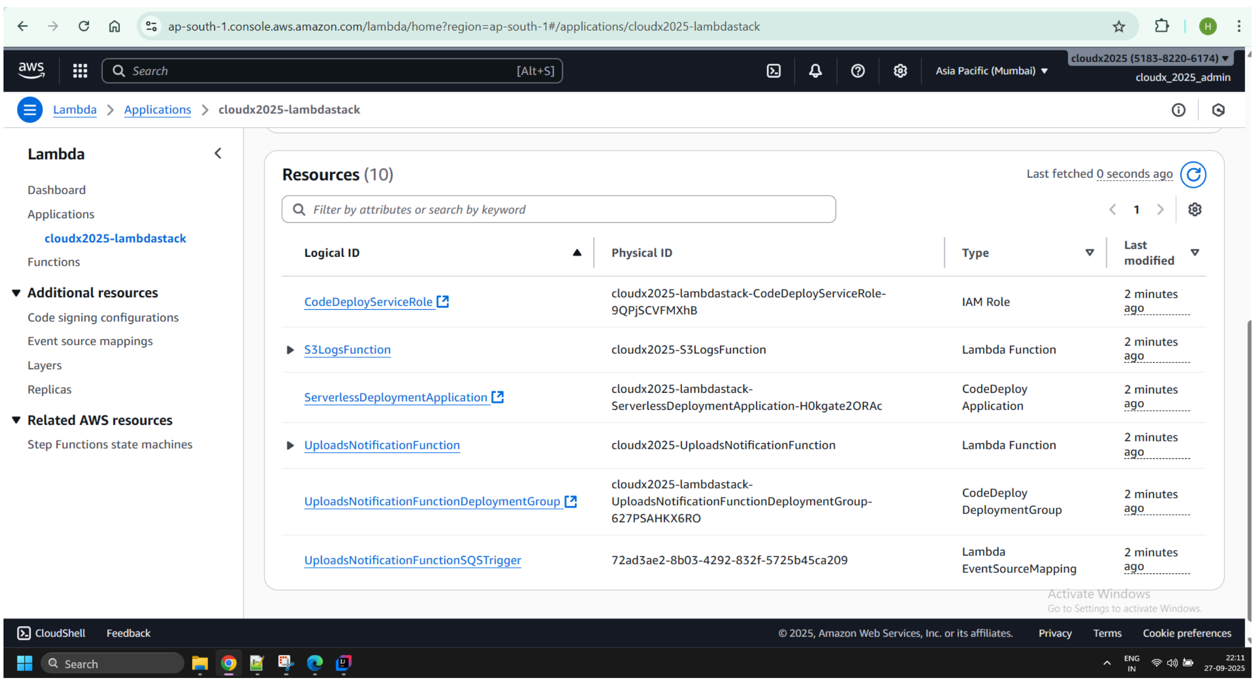
ap-south-1.console.aws.amazon.com/lambda/home?region=ap-south-1#/applications/cloudx2025-lambdastack
Search
[Alt+S]
Asia Pacific (Mumbai)
cloudx2025 (5183-8220-6174)
cloudx_2025_admin
Lambda > Applications > cloudx2025-lambdastack
Lambda
Dashboard
Applications
cloudx2025-lambdastack
Functions
Additional resources
Code signing configurations
Event source mappings
Layers
Replicas
Related AWS resources
Step Functions state machines
Resources (10)
Last fetched 0 seconds ago
Filter by attributes or search by keyword
Logical ID
Physical ID
Type
Last modified
CodeDeployServiceRole
cloudx2025-lambdastack-CodeDeployServiceRole-9QPjSCVFMXhB
IAM Role
2 minutes ago
S3LogsFunction
cloudx2025-S3LogsFunction
Lambda Function
2 minutes ago
ServerlessDeploymentApplication
cloudx2025-lambdastack-ServerlessDeploymentApplication-H0kgate2ORAc
CodeDeploy Application
2 minutes ago
UploadsNotificationFunction
cloudx2025-UploadsNotificationFunction
Lambda Function
2 minutes ago
UploadsNotificationFunctionDeploymentGroup
cloudx2025-lambdastack-UploadsNotificationFunctionDeploymentGroup-627PSAHKX6RO
CodeDeploy DeploymentGroup
2 minutes ago
UploadsNotificationFunctionSQSTrigger
72ad3ae2-8b03-4292-832f-5725b45ca209
Lambda EventSourceMapping
2 minutes ago
CloudShell
Feedback
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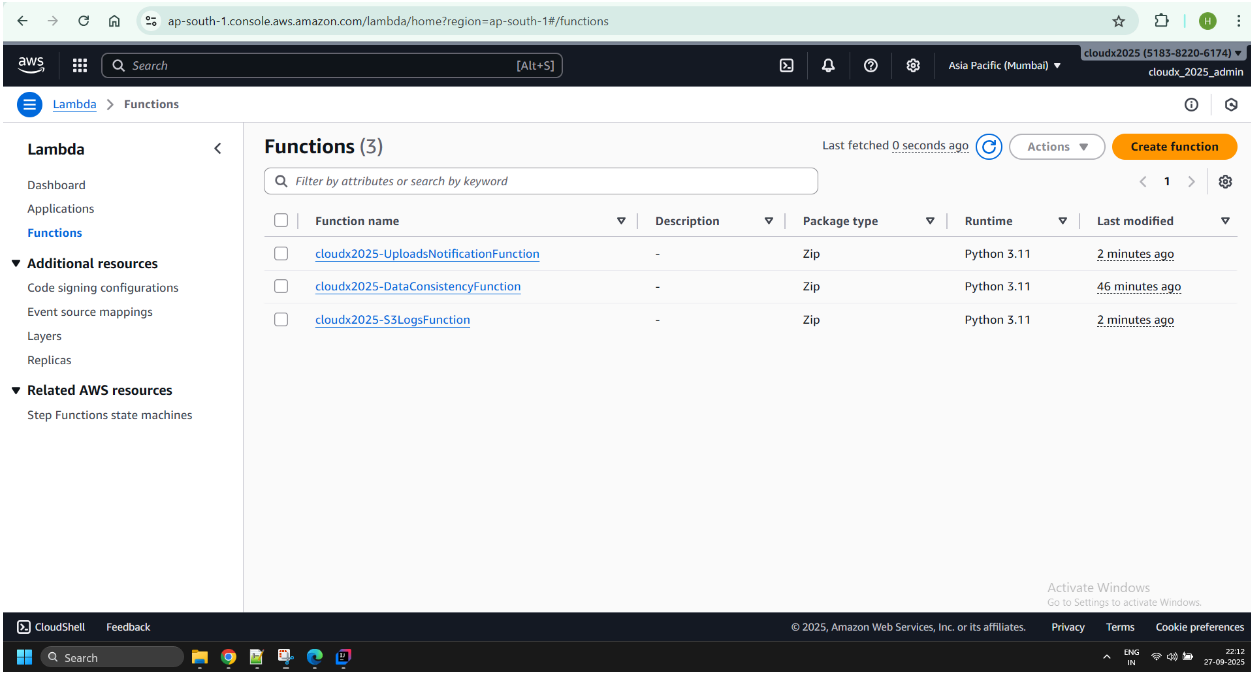
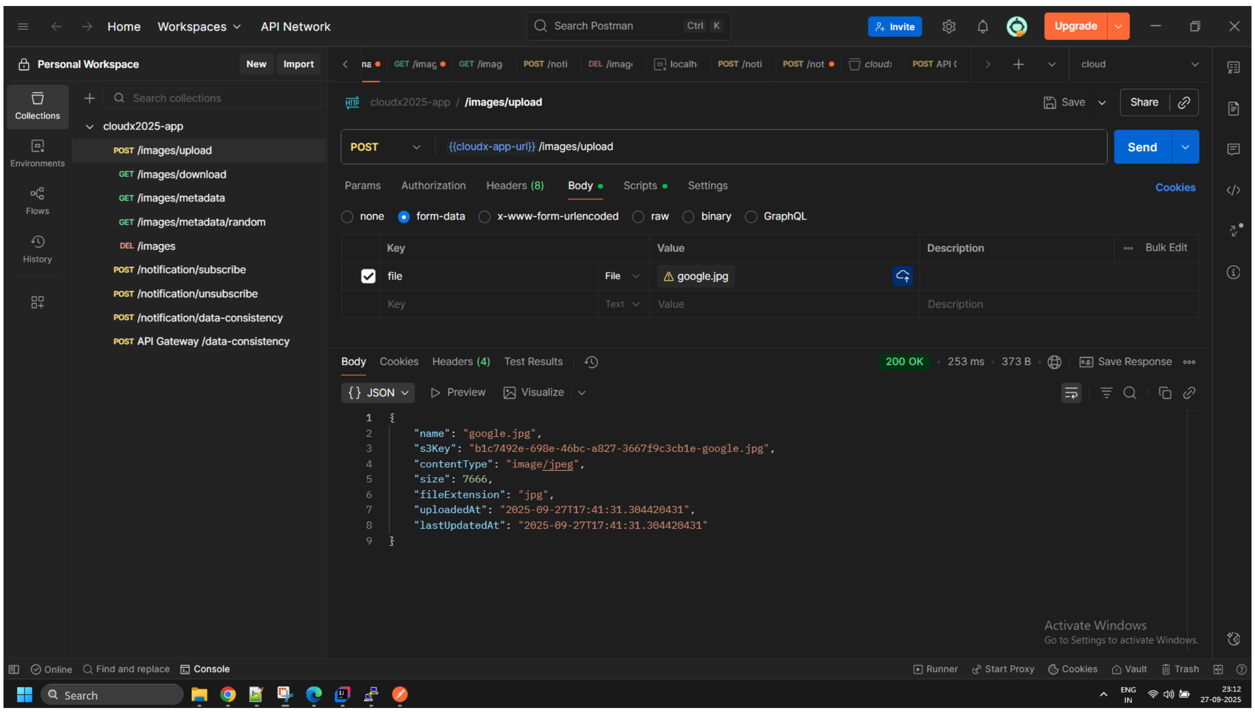
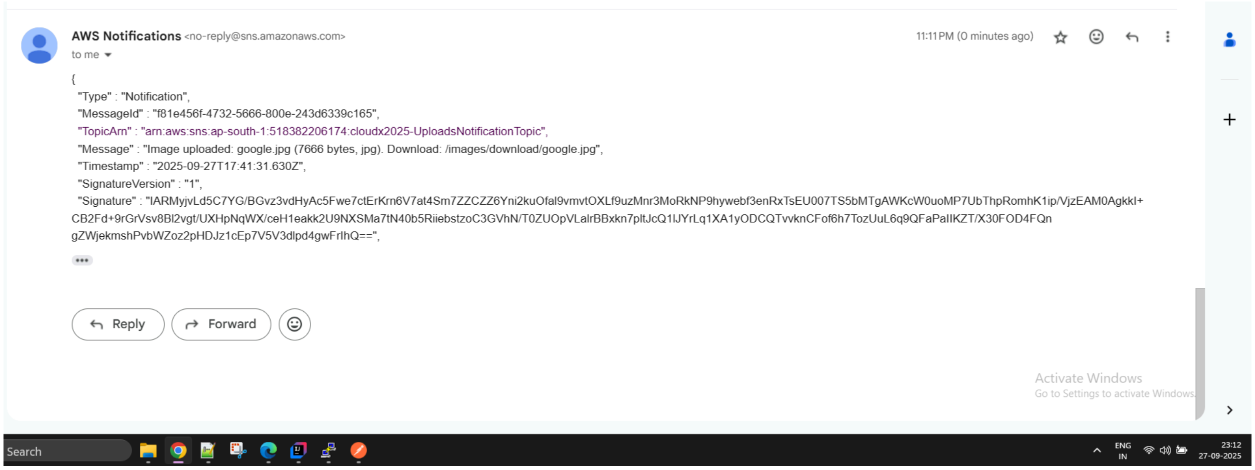

ap-south-1.console.aws.amazon.com/lambda/home?region=ap-south-1#/functions
Lambda > Functions
Functions (3)
Last fetched 0 seconds ago
Actions
Create function
Filter by attributes or search by keyword
Function name | Description | Package type | Runtime | Last modified
cloudx2025-UploadsNotificationFunction - Zip Python 3.11 2 minutes ago
cloudx2025-DataConsistencyFunction - Zip Python 3.11 46 minutes ago
cloudx2025-S3LogsFunction - Zip Python 3.11 2 minutes ago
Dashboard
Applications
Functions
Additional resources
Code signing configurations
Event source mappings
Layers
Replicas
Related AWS resources
Step Functions state machines


cloudx2025-app / /images/upload
POST {{cloudx-app-url}} /images/upload
Send
Body form-data
file google.jpg
200 OK 253 ms 373 B
{
"name": "google.jpg",
"s3Key": "b1c7492e-698e-46bc-a827-3667f9c3cb1e-google.jpg",
"contentType": "image/jpeg",
"size": 7666,
"fileExtension": "jpg",
"uploadedAt": "2025-09-27T17:41:31.304420431",
"lastUpdatedAt": "2025-09-27T17:41:31.304420431"
}
POST /images/upload
GET /images/download
GET /images/metadata
GET /images/metadata/random
DEL /images
POST /notification/subscribe
POST /notification/unsubscribe
POST /notification/data-consistency
POST API Gateway /data-consistency



Network stack

```yaml
Parameters:
 ProjectName:
   Type: String
 BucketSuffix:
   Type: String
   Description: "Suffix for the S3 bucket name"
 DBUsername:
   Type: String
   NoEcho: true
   Description: "Master username for RDS"
 DBPassword:
   Type: String
   NoEcho: true
   Description: "Master password for RDS"
 DBName:
   Type: String
   Default: cloudx2025
   Description: "Name of the database to create"

Resources:
 S3Bucket:
   Type: AWS::S3::Bucket
   Properties:
     BucketName: !Sub "cloudx2025-${BucketSuffix}"

 VPC:
   Type: AWS::EC2::VPC
   Properties:
     CidrBlock: 10.0.0.0/16
     EnableDnsHostnames: true
     EnableDnsSupport: true
     Tags:
       - Key: Name
```

```yaml
        Value: !Sub "${ProjectName}-VPC"

InternetGateway:
  Type: AWS::EC2::InternetGateway
  Properties:
    Tags:
      - Key: Name
        Value: !Sub "${ProjectName}-IGW"

AttachGateway:
  Type: AWS::EC2::VPCGatewayAttachment
  Properties:
    VpcId: !Ref VPC
    InternetGatewayId: !Ref InternetGateway

PublicSubnetA:
  Type: AWS::EC2::Subnet
  Properties:
    VpcId: !Ref VPC
    CidrBlock: 10.0.1.0/24
    AvailabilityZone: !Select [0, !GetAZs '']
    MapPublicIpOnLaunch: true
    Tags:
      - Key: Name
        Value: !Sub "${ProjectName}-PublicSubnet-A"

PublicSubnetB:
  Type: AWS::EC2::Subnet
  Properties:
    VpcId: !Ref VPC
    CidrBlock: 10.0.2.0/24
    AvailabilityZone: !Select [1, !GetAZs '']
    MapPublicIpOnLaunch: true
    Tags:
      - Key: Name
        Value: !Sub "${ProjectName}-PublicSubnet-B"

PrivateSubnetA:
  Type: AWS::EC2::Subnet
  Properties:
    VpcId: !Ref VPC
    CidrBlock: 10.0.3.0/24
    AvailabilityZone: !Select [0, !GetAZs '']  # This will be ap-south-1a
    MapPublicIpOnLaunch: false
    Tags:
      - Key: Name
        Value: !Sub "${ProjectName}-PrivateSubnetA"

PrivateSubnetB:
```

```yaml
    Type: AWS::EC2::Subnet
    Properties:
      VpcId: !Ref VPC
      CidrBlock: 10.0.4.0/24
      AvailabilityZone: !Select [1, !GetAZs '']  # Changed to index 1 for
ap-south-1b
      MapPublicIpOnLaunch: false
      Tags:
        - Key: Name
          Value: !Sub "${ProjectName}-PrivateSubnetB"

  PublicRouteTable:
    Type: AWS::EC2::RouteTable
    Properties:
      VpcId: !Ref VPC
      Tags:
        - Key: Name
          Value: !Sub "${ProjectName}-PublicRT"

  PublicRoute:
    Type: AWS::EC2::Route
    DependsOn: AttachGateway
    Properties:
      RouteTableId: !Ref PublicRouteTable
      DestinationCidrBlock: 0.0.0.0/0
      GatewayId: !Ref InternetGateway

  PublicSubnetARouteTableAssociation:
    Type: AWS::EC2::SubnetRouteTableAssociation
    Properties:
      SubnetId: !Ref PublicSubnetA
      RouteTableId: !Ref PublicRouteTable

  PublicSubnetBRouteTableAssociation:
    Type: AWS::EC2::SubnetRouteTableAssociation
    Properties:
      SubnetId: !Ref PublicSubnetB
      RouteTableId: !Ref PublicRouteTable

  PrivateRouteTable:
    Type: AWS::EC2::RouteTable
    Properties:
      VpcId: !Ref VPC
      Tags:
        - Key: Name
          Value: !Sub "${ProjectName}-PrivateRT"

  PrivateSubnetARouteTableAssociation:
    Type: AWS::EC2::SubnetRouteTableAssociation
```

```yaml
    Properties:
      SubnetId: !Ref PrivateSubnetA
      RouteTableId: !Ref PrivateRouteTable

  PrivateSubnetBRouteTableAssociation:
    Type: AWS::EC2::SubnetRouteTableAssociation
    Properties:
      SubnetId: !Ref PrivateSubnetB
      RouteTableId: !Ref PrivateRouteTable

  S3VpcEndpoint:
    Type: AWS::EC2::VPCEndpoint
    Properties:
      VpcEndpointType: Gateway
      ServiceName: !Sub "com.amazonaws.${AWS::Region}.s3"
      VpcId: !Ref VPC
      RouteTableIds:
        - !Ref PrivateRouteTable

  DBSubnetGroup:
    Type: AWS::RDS::DBSubnetGroup
    Properties:
      DBSubnetGroupDescription: !Sub "${ProjectName} RDS subnet group"
      SubnetIds:
        - !Ref PrivateSubnetA
        - !Ref PrivateSubnetB
      DBSubnetGroupName: !Sub "${ProjectName}-db-subnet"

  AppSecurityGroup:
    Type: AWS::EC2::SecurityGroup
    Properties:
      GroupDescription: "Allow app traffic from ALB and SSH access"
      VpcId: !Ref VPC
      SecurityGroupIngress:
        - IpProtocol: tcp
          FromPort: 22
          ToPort: 22
          CidrIp: 103.41.39.252/32
      Tags:
        - Key: Name
          Value: !Sub "${ProjectName}-AppSecurityGroup"

  RdsSecurityGroup:
    Type: AWS::EC2::SecurityGroup
    Properties:
      GroupDescription: "Allow MySQL from app"
      VpcId: !Ref VPC
      SecurityGroupIngress:
        - IpProtocol: tcp
```

```yaml
          FromPort: 3306
          ToPort: 3306
          SourceSecurityGroupId: !Ref AppSecurityGroup

  MyDB:
    Type: AWS::RDS::DBInstance
    Properties:
      Engine: mysql
      DBInstanceClass: db.t3.micro
      AllocatedStorage: 20
      DBName: !Ref DBName
      MasterUsername: !Ref DBUsername
      MasterUserPassword: !Ref DBPassword
      PubliclyAccessible: false
      MultiAZ: true
      BackupRetentionPeriod: 0  # Disable automated backups
      VPCSecurityGroups:
        - !Ref RdsSecurityGroup
      DBSubnetGroupName: !Ref DBSubnetGroup
      DeletionProtection: false

Outputs:
  VpcId:
    Value: !Ref VPC
    Export:
      Name: !Sub "${ProjectName}-VpcId"
  PublicSubnetAId:
    Value: !Ref PublicSubnetA
    Export:
      Name: !Sub "${ProjectName}-PublicSubnetAId"
  PublicSubnetBId:
    Value: !Ref PublicSubnetB
    Export:
      Name: !Sub "${ProjectName}-PublicSubnetBId"
  PrivateSubnetAId:
    Value: !Ref PrivateSubnetA
    Export:
      Name: !Sub "${ProjectName}-PrivateSubnetAId"
  PrivateSubnetBId:
    Value: !Ref PrivateSubnetB
    Export:
      Name: !Sub "${ProjectName}-PrivateSubnetBId"
  S3BucketName:
    Value: !Ref S3Bucket
    Export:
      Name: !Sub "${ProjectName}-S3BucketName"
  RDSEndpoint:
    Value: !GetAtt MyDB.Endpoint.Address
    Export:
```

```yaml
      Name: !Sub "${ProjectName}-RDSEndpoint"
  RDSInstanceId:
    Value: !Ref MyDB
    Export:
      Name: !Sub "${ProjectName}-RDSInstanceId"
  RDSUsername:
    Value: !Ref DBUsername
    Export:
      Name: !Sub "${ProjectName}-RDSUsername"
  AppSecurityGroupId:
    Value: !Ref AppSecurityGroup
    Export:
      Name: !Sub "${ProjectName}-AppSecurityGroupId"
  RDSDBName:
    Value: !Ref DBName
    Export:
      Name: !Sub "${ProjectName}-RDSDBName"
  RdsSecurityGroupId:
    Value: !Ref RdsSecurityGroup
    Export:
      Name: !Sub "${ProjectName}-RdsSecurityGroupId"
```

App stack

```yaml
AWSTemplateFormatVersion: '2010-09-09'
Transform: AWS::Serverless-2016-10-31
Description: Application stack with Lambda functions

Parameters:
  ProjectName:
    Type: String
  DeploymentBucket:
    Type: String
    Description: "S3 bucket name containing deployment artifacts (JAR, Lambda
code, layers)"
  AppJarKey:
    Type: String
    Description: "S3 key for the application JAR file"
  DataConsistencyCodeKey:
    Type: String
    Default: "data_consistency.zip"
    Description: "S3 key for the Data Consistency Lambda function code"
  LayerKey:
    Type: String
    Description: "S3 key for the Lambda layer zip file"
  RDSHost:
    Type: String
    Description: "RDS endpoint from network stack"
  DBUsername:
```

```yaml
    Type: String
    Description: "Database username"
    NoEcho: true
  DBPassword:
    Type: String
    Description: "Database password"
    NoEcho: true
  S3BucketName:
    Type: String
    Description: "S3 bucket name from network stack"
  DBName:
    Type: String
    Description: "Database name from network stack"

Resources:
  AppSecurityGroupIngress:
    Type: AWS::EC2::SecurityGroupIngress
    Properties:
      GroupId:
        Fn::ImportValue: !Sub "${ProjectName}-AppSecurityGroupId"
      IpProtocol: tcp
      FromPort: 8080
      ToPort: 8080
      SourceSecurityGroupId: !Ref AlbSecurityGroup

  AlbSecurityGroup:
    Type: AWS::EC2::SecurityGroup
    Properties:
      GroupDescription: "Security group for ALB"
      VpcId:
        Fn::ImportValue: !Sub "${ProjectName}-VpcId"
      SecurityGroupIngress:
        - IpProtocol: tcp
          FromPort: 80
          ToPort: 80
          CidrIp: 0.0.0.0/0
      Tags:
        - Key: Name
          Value: !Sub "${ProjectName}-ALBSecurityGroup"

  AppLaunchTemplate:
    Type: AWS::EC2::LaunchTemplate
    Properties:
      LaunchTemplateData:
        ImageId: ami-0b982602dbb32c5bd
        InstanceType: t3.micro
        KeyName: cloudx_ec2_keypair
        IamInstanceProfile:
          Name: cloudx2025_FullAccessRoleS3
```

```
      SecurityGroupIds:
        - Fn::ImportValue: !Sub "${ProjectName}-AppSecurityGroupId"
      UserData:
        Fn::Base64: !Sub |
          #!/bin/bash
          yum install -y aws-cli

          # Install Java 17
          amazon-linux-extras enable java-17-amazon-corretto
          yum clean metadata
          yum install -y java-17-amazon-corretto

          # Download and start the application
          aws s3 cp s3://${DeploymentBucket}/${AppJarKey}
/home/ec2-user/cloudx2025-app.jar

          # Create application service file with properly escaped environment
variables
          cat << EOF > /etc/systemd/system/cloudx-app.service
          [Unit]
          Description=CloudX Application
          After=network.target

          [Service]
          User=ec2-user
          WorkingDirectory=/home/ec2-user
          Environment=DB_HOST=${RDSHost}
          Environment=DB_USER=${DBUsername}
          Environment=DB_PASS=${DBPassword}
          Environment=DB_NAME=${DBName}
          Environment=S3_BUCKET=${S3BucketName}
          Environment=SERVER_PORT=8080
          Environment=PROJECT_NAME=${ProjectName}

Environment=UPLOADS_NOTIFICATION_QUEUE_URL=${UploadsNotificationQueue}

Environment=UPLOADS_NOTIFICATION_TOPIC_ARN=${UploadsNotificationTopic}
          ExecStart=/usr/bin/java -jar cloudx2025-app.jar
          SuccessExitStatus=143
          TimeoutStopSec=10
          Restart=on-failure
          RestartSec=5

          [Install]
          WantedBy=multi-user.target
          EOF

          # Log the environment variables for debugging
          echo "Environment variables being used:" > /home/ec2-user/env.log
```

```
          echo "DB_HOST=${RDSHost}" >> /home/ec2-user/env.log
          echo "DB_USER=${DBUsername}" >> /home/ec2-user/env.log
          echo "DB_NAME=${DBName}" >> /home/ec2-user/env.log
          echo "S3_BUCKET=${S3BucketName}" >> /home/ec2-user/env.log
          echo "SERVER_PORT=8080" >> /home/ec2-user/env.log

          # Set permissions and start service
          chmod 644 /etc/systemd/system/cloudx-app.service
          systemctl daemon-reload
          systemctl enable cloudx-app
          systemctl start cloudx-app

          # Save logs to persistent location
          ln -sf /var/log/messages /home/ec2-user/system.log
          touch /home/ec2-user/app.log
          chown ec2-user:ec2-user /home/ec2-user/app.log
          journalctl -u cloudx-app -f > /home/ec2-user/app.log &

  AppAutoScalingGroup:
    Type: AWS::AutoScaling::AutoScalingGroup
    Properties:
      VPCZoneIdentifier:
        - Fn::ImportValue: !Sub "${ProjectName}-PublicSubnetAId"
        - Fn::ImportValue: !Sub "${ProjectName}-PublicSubnetBId"
      LaunchTemplate:
        LaunchTemplateId: !Ref AppLaunchTemplate
        Version: !GetAtt AppLaunchTemplate.LatestVersionNumber
      MinSize: "1"
      MaxSize: "3"
      DesiredCapacity: "1"
      TargetGroupARNs:
        - !Ref AppTargetGroup
      Tags:
        - Key: Name
          Value: !Sub "${ProjectName}-AppInstance"
          PropagateAtLaunch: true

  AppTargetGroup:
    Type: AWS::ElasticLoadBalancingV2::TargetGroup
    Properties:
      VpcId:
        Fn::ImportValue: !Sub "${ProjectName}-VpcId"
      Port: 8080
      Protocol: HTTP
      TargetType: instance
      HealthCheckProtocol: HTTP
      HealthCheckPort: "8080"
      HealthCheckPath: "/"
      HealthCheckIntervalSeconds: 15
```

```yaml
      HealthCheckTimeoutSeconds: 5
      HealthyThresholdCount: 2
      UnhealthyThresholdCount: 5
      Matcher:
        HttpCode: 200
      Name: !Sub "${ProjectName}-tg"

  AppLoadBalancer:
    Type: AWS::ElasticLoadBalancingV2::LoadBalancer
    Properties:
      Subnets:
        - Fn::ImportValue: !Sub "${ProjectName}-PublicSubnetAId"
        - Fn::ImportValue: !Sub "${ProjectName}-PublicSubnetBId"
      SecurityGroups:
        - !Ref AlbSecurityGroup
      Tags:
        - Key: Name
          Value: !Sub "${ProjectName}-LoadBalancer"

  AppListener:
    Type: AWS::ElasticLoadBalancingV2::Listener
    Properties:
      DefaultActions:
        - Type: forward
          TargetGroupArn: !Ref AppTargetGroup
      LoadBalancerArn: !Ref AppLoadBalancer
      Port: 80
      Protocol: HTTP

  UploadsNotificationQueue:
    Type: AWS::SQS::Queue
    Properties:
      QueueName: !Sub "${ProjectName}-UploadsNotificationQueue"

  UploadsNotificationTopic:
    Type: AWS::SNS::Topic
    Properties:
      TopicName: !Sub "${ProjectName}-UploadsNotificationTopic"

  DataConsistencyLambdaRole:
    Type: AWS::IAM::Role
    Properties:
      RoleName: !Sub "${ProjectName}-DataConsistencyLambdaRole"
      AssumeRolePolicyDocument:
        Version: "2012-10-17"
        Statement:
          - Effect: Allow
            Principal:
              Service: lambda.amazonaws.com
```

```yaml
          Action: sts:AssumeRole
      ManagedPolicyArns:
        - arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole
        - arn:aws:iam::aws:policy/service-role/AWSLambdaVPCAccessExecutionRole
        - arn:aws:iam::aws:policy/AmazonS3FullAccess
        - arn:aws:iam::aws:policy/AmazonRDSFullAccess

  DataConsistencySecurityGroup:
    Type: AWS::EC2::SecurityGroup
    Properties:
      GroupDescription: "Security group for Data Consistency Lambda"
      VpcId:
        Fn::ImportValue: !Sub "${ProjectName}-VpcId"

  RdsSecurityGroupIngressFromLambda:
    Type: AWS::EC2::SecurityGroupIngress
    Properties:
      GroupId:
        Fn::ImportValue: !Sub "${ProjectName}-RdsSecurityGroupId"
      IpProtocol: tcp
      FromPort: 3306
      ToPort: 3306
      SourceSecurityGroupId: !Ref DataConsistencySecurityGroup

  LambdaInvokeInlinePolicy:
    Type: AWS::IAM::Policy
    Properties:
      PolicyName: !Sub "${ProjectName}-LambdaInvokeInlinePolicy"
      PolicyDocument:
        Version: "2012-10-17"
        Statement:
          - Effect: Allow
            Action: lambda:InvokeFunction
            Resource: !GetAtt DataConsistencyFunction.Arn
      Roles:
        - cloudx2025_FullAccessRoleS3

  DataConsistencyLayer:
    Type: AWS::Lambda::LayerVersion
    Properties:
      LayerName: !Sub "${ProjectName}-DataConsistencyLayer"
      CompatibleRuntimes:
        - python3.11
      Content:
        S3Bucket: !Ref DeploymentBucket
        S3Key: !Ref LayerKey

  DataConsistencyFunction:
    Type: AWS::Serverless::Function
```

```
    Properties:
      FunctionName: !Sub "${ProjectName}-DataConsistencyFunction"
      Runtime: python3.11
      Handler: data_consistency.lambda_handler
      Role: !GetAtt DataConsistencyLambdaRole.Arn
      Layers:
        - !Ref DataConsistencyLayer
      VpcConfig:
        SecurityGroupIds:
          - !Ref DataConsistencySecurityGroup
        SubnetIds:
          - Fn::ImportValue: !Sub "${ProjectName}-PrivateSubnetAId"
          - Fn::ImportValue: !Sub "${ProjectName}-PrivateSubnetBId"
      Environment:
        Variables:
          DB_HOST: !Ref RDSHost
          DB_USER: !Ref DBUsername
          DB_PASS: !Ref DBPassword
          DB_NAME: !Ref DBName
          S3_BUCKET: !Ref S3BucketName
      CodeUri:
        Bucket: !Ref DeploymentBucket
        Key: !Ref DataConsistencyCodeKey
      Events:
        DataConsistencySchedule:
          Type: Schedule
          Properties:
            Schedule: rate(20 minutes)
            Name: !Sub "${ProjectName}-DataConsistencySchedule"
            Description: "Schedule to run data consistency check every 20
minutes"
            Input: |
              {
                "version": "0",
                "id": "scheduled-event",
                "detail-type": "ScheduledEvent",
                "source": "aws.events",
                "detail": {}
              }
        DataConsistencyApi:
          Type: Api
          Properties:
            Path: /data-consistency
            Method: post

Outputs:
 AppAutoScalingGroupName:
    Value: !Ref AppAutoScalingGroup
    Export:
```

```yaml
      Name: !Sub "${ProjectName}-AppAutoScalingGroupName"
  AppLoadBalancerDNS:
    Value: !GetAtt AppLoadBalancer.DNSName
    Export:
      Name: !Sub "${ProjectName}-AppLoadBalancerDNS"
  UploadsNotificationQueueArn:
    Description: "ARN of the uploads notification SQS queue"
    Value: !GetAtt UploadsNotificationQueue.Arn
    Export:
      Name: !Sub "${ProjectName}-UploadsNotificationQueueArn"
  UploadsNotificationTopicArn:
    Description: "ARN of the uploads notification SNS topic"
    Value: !Ref UploadsNotificationTopic
    Export:
      Name: !Sub "${ProjectName}-UploadsNotificationTopicArn"
  UploadsNotificationQueueUrl:
    Description: "URL of the uploads notification SQS queue"
    Value: !Ref UploadsNotificationQueue
    Export:
      Name: !Sub "${ProjectName}-UploadsNotificationQueueUrl"
```

SAM template / Lambda stack

```yaml
AWSTemplateFormatVersion: '2010-09-09'
Transform: AWS::Serverless-2016-10-31
Description: SAM template for <ProjectName>-UploadsNotificationFunction

Parameters:
  ProjectName:
    Type: String
    Description: "Project name for resource naming"
  SnsTopicArn:
    Type: String
    Description: "ARN of the SNS topic"
  SqsQueueArn:
    Type: String
    Description: "ARN of the SQS queue"
  SqsQueueUrl:
    Type: String
    Description: "URL of the SQS queue"
  S3BucketName:
    Type: String
    Description: "S3 bucket name for S3LogsFunction"

Resources:
  UploadsNotificationLambdaRole:
    Type: AWS::IAM::Role
    Properties:
      RoleName: !Sub "${ProjectName}-UploadsNotificationLambdaRole"
      AssumeRolePolicyDocument:
```

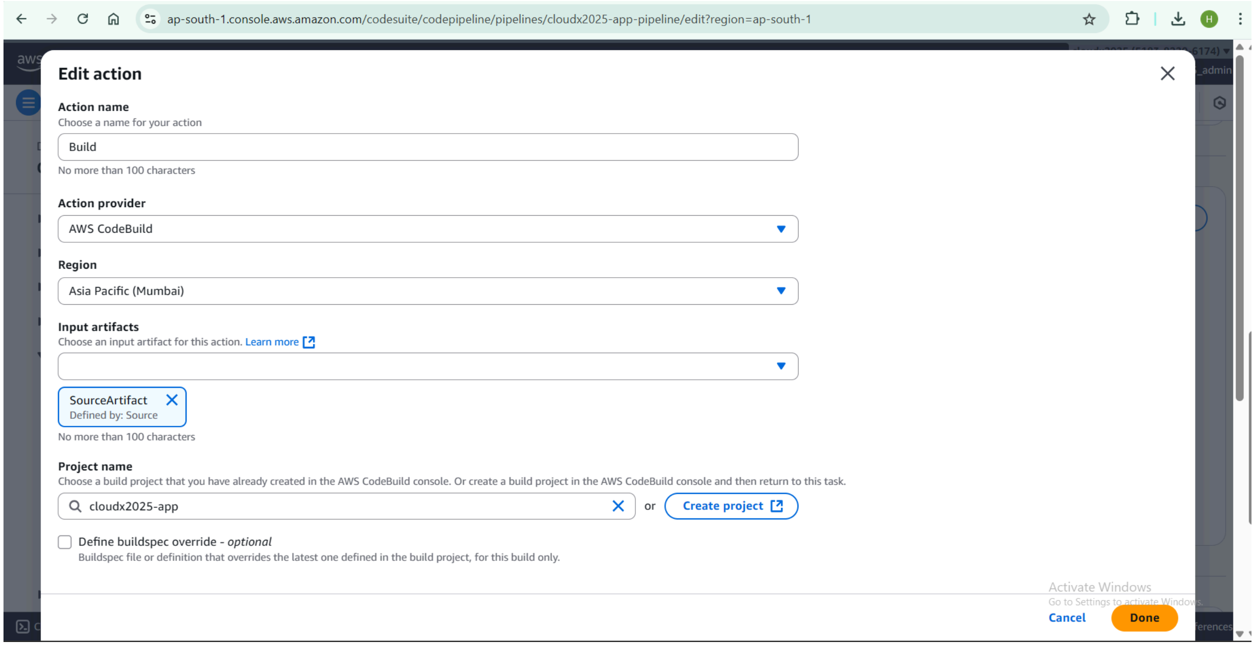
```yaml
          Version: "2012-10-17"
          Statement:
            - Effect: Allow
              Principal:
                Service: lambda.amazonaws.com
              Action: sts:AssumeRole
        ManagedPolicyArns:
          - arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole
        Policies:
          - PolicyName: SNSPublishPolicy
            PolicyDocument:
              Version: "2012-10-17"
              Statement:
                - Effect: Allow
                  Action: sns:Publish
                  Resource: !Ref SnsTopicArn
          - PolicyName: SQSReceivePolicy
            PolicyDocument:
              Version: "2012-10-17"
              Statement:
                - Effect: Allow
                  Action:
                    - sqs:ReceiveMessage
                    - sqs:DeleteMessage
                    - sqs:GetQueueAttributes
                  Resource: !Ref SqsQueueArn

  UploadsNotificationFunction:
    Type: AWS::Serverless::Function
    Properties:
      FunctionName: !Sub "${ProjectName}-UploadsNotificationFunction"
      Runtime: python3.11
      Handler: uploads_notification.lambda_handler
      Role: !GetAtt UploadsNotificationLambdaRole.Arn
      CodeUri: ./uploads_notification/
      Environment:
        Variables:
          SNS_TOPIC_ARN: !Ref SnsTopicArn
      Events:
        SQSTrigger:
          Type: SQS
          Properties:
            Queue: !Ref SqsQueueArn
            BatchSize: 10
      DeploymentPreference:
        Type: AllAtOnce
        Alarms:
          - !Sub "${ProjectName}-UploadsNotificationFunction-Alarm"
        Hooks:
```

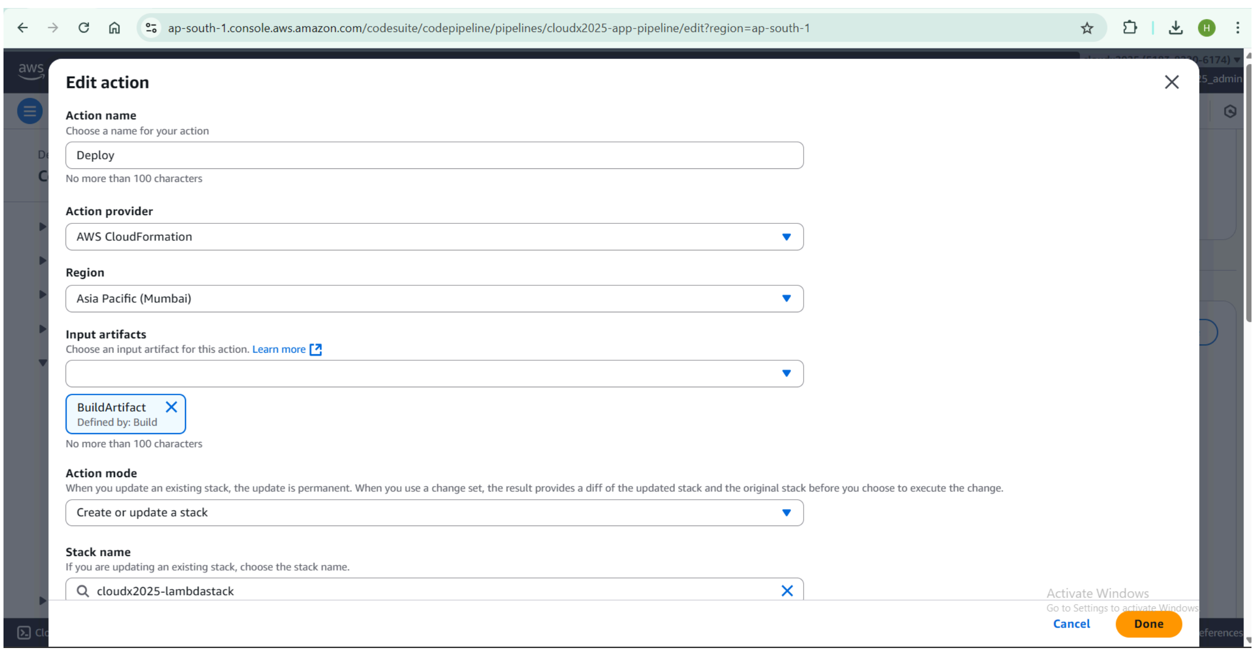
```yaml
          PreTraffic: !Sub "${ProjectName}-PreTrafficHook"
          PostTraffic: !Sub "${ProjectName}-PostTrafficHook"
      Timeout: 30
      AutoPublishAlias: live

  # Optional: S3 Logs Lambda function for testing
  S3LogsFunction:
    Type: AWS::Serverless::Function
    Properties:
      FunctionName: !Sub "${ProjectName}-S3LogsFunction"
      Runtime: python3.11
      Handler: s3_logs.lambda_handler
      CodeUri: ./s3_logs/
      Policies:
        - S3ReadPolicy:
            BucketName: !Ref S3BucketName
      Timeout: 15

Outputs:
  UploadsNotificationFunctionArn:
    Description: "ARN of the UploadsNotification Lambda function"
    Value: !GetAtt UploadsNotificationFunction.Arn
  S3LogsFunctionArn:
    Description: "ARN of the S3Logs Lambda function"
    Value: !GetAtt S3LogsFunction.Arn
```

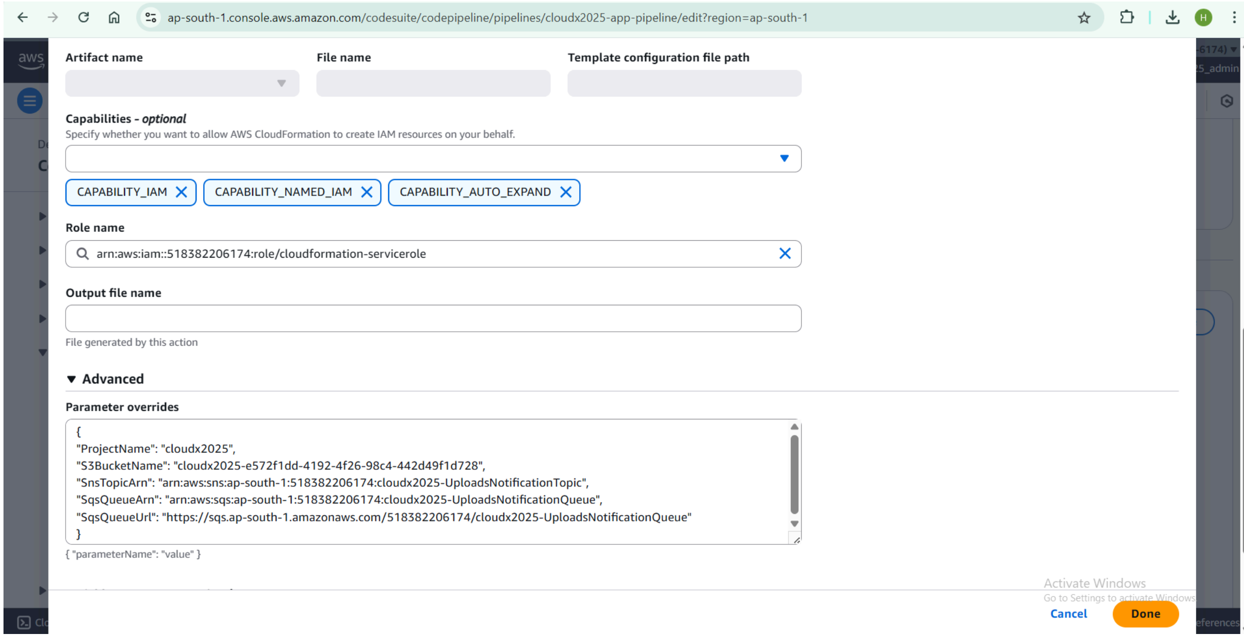
Sub-task 2 – Build CI/CD pipeline

1. We need to create a CodeConnection that will be used in the configuration of CodeBuild and CodePipeline. At this stage, we need to select a provider, such as GitHub using this tutorial or GitLab using this tutorial. The following instructions are tailored for the GitHub provider.
2. Following this tutorial create a CodeBuild spec, which would:
    - provide instructions for testing the code (optionally)
    - build the executable artifact(for example jar file for Java)
    - package the code for the Lambda function using sam package CLI command.
    - specify the generated output template file as build artifact in the artifacts section for further using it for deploy.
    - Add this created CodeBuild spec in your repository

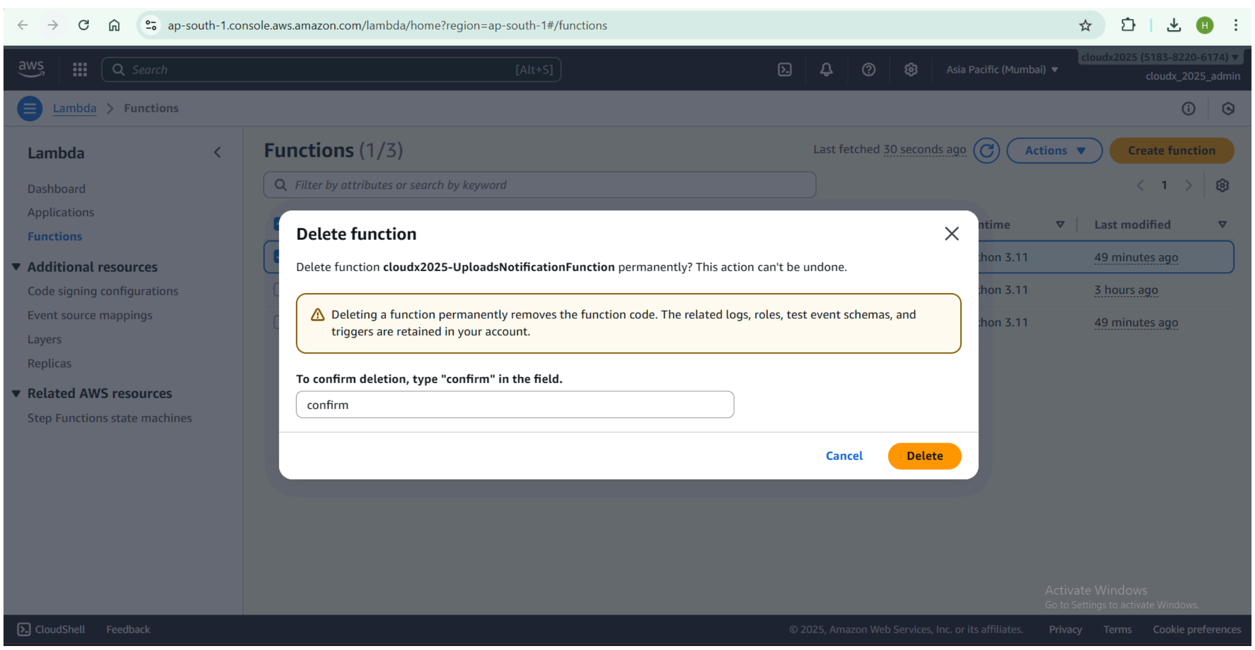
    - Important: don’t use sam deploy cli command here. The *<ProjectName>-UploadsNotificationFunction* Lambda will be deployed through CodeDeploy service.
3. Create a CodeBuild project using AWS console:
    - Choose your source provider (GitHub or GitLab). Navigate to the 'Credential' section and click on 'Manage account credentials.' In the 'Manage default source credential' window that appears, select 'Credential Type' as either 'GitHub App' or 'GitLab.' Then, choose the CodeConnection you created in the #1 step.
    - Choose your own repository, which already contains the CodeBuild spec, as the source provider. Ensure that you have previously created this CodeBuild spec and added it to your repository in a prior step.
    - configure environment for building the artifact
    - specify S3 bucket for storing the build artifact
    - Manually start the build. Verify that output template file, generated by sam package command, was stored in S3. In the CodeUri section of this file, you could find the link to the packaged Lambda code, stored in S3.
4. Note. If you see the error 'Webhook creation failed' while creating a CodeBuild project, click 'Edit' for that CodeBuild project, check the 'Rebuild every time a code change is pushed to this repository' option in the 'webhook' section again, and save. This should resolve the issue.
5. Create a CodePipeline project and configure:
    - In 'Creation options' select 'Build custom pipeline'.
    - Source Stage - GitHub/GitLab - automatically runs after commit into your repo from and pulls all your sources
    - Build Stage - Code Build - uses your source code and runs SAM package to upload lambda source/package to S3 and generates CloudFormation output template. You could reuse already create build project from the previous step
    - Deploy Stage - Cloud Formation as action provider - uses output template from second step to deploy lambda


ap-south-1.console.aws.amazon.com/codesuite/codepipeline/pipelines/cloudx2025-app-pipeline/edit?region=ap-south-1
Edit action
Action name
Choose a name for your action
Build
No more than 100 characters
Action provider
AWS CodeBuild
Region
Asia Pacific (Mumbai)
Input artifacts
Choose an input artifact for this action. Learn more
SourceArtifact
Defined by: Source
No more than 100 characters
Project name
Choose a build project that you have already created in the AWS CodeBuild console. Or create a build project in the AWS CodeBuild console and then return to this task.
cloudx2025-app
or
Create project
Define buildspec override - optional
Buildspec file or definition that overrides the latest one defined in the build project, for this build only.
Activate Windows
Go to Settings to activate Windows.
Cancel
Done

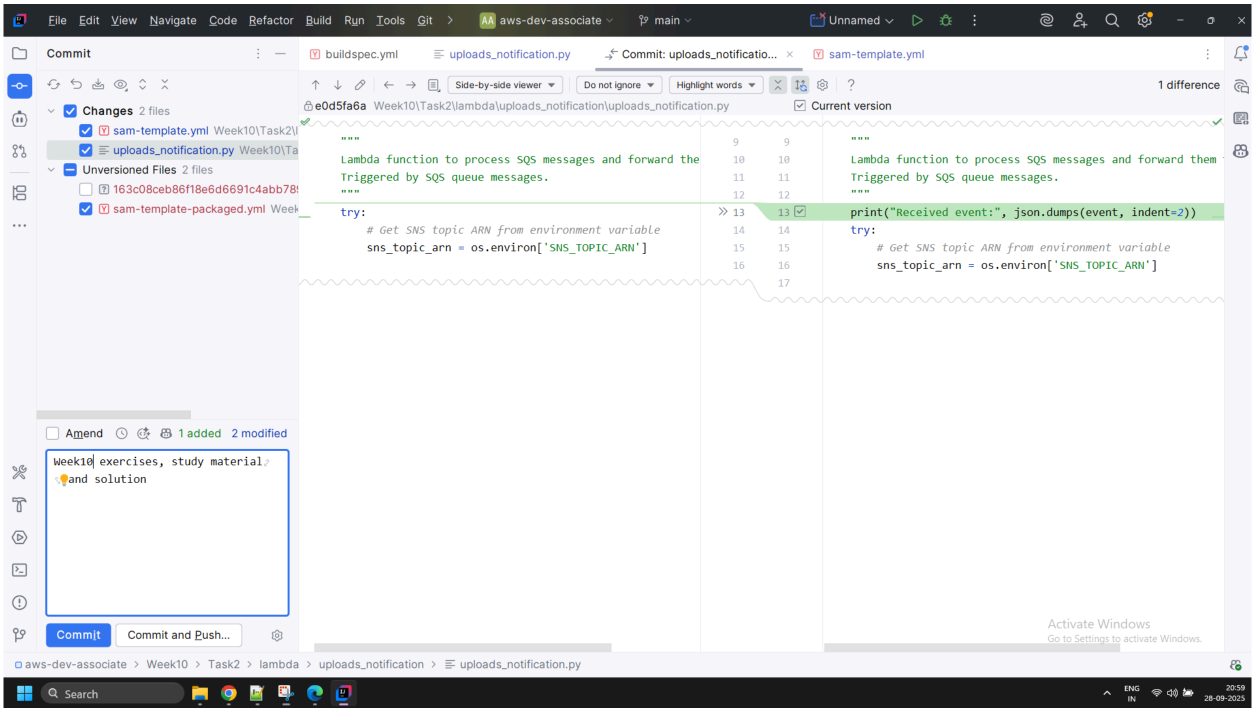

ap-south-1.console.aws.amazon.com/codesuite/codepipeline/pipelines/cloudx2025-app-pipeline/edit?region=ap-south-1
Edit action
Action name
Choose a name for your action
Deploy
No more than 100 characters
Action provider
AWS CloudFormation
Region
Asia Pacific (Mumbai)
Input artifacts
Choose an input artifact for this action. Learn more
BuildArtifact
Defined by: Build
No more than 100 characters
Action mode
When you update an existing stack, the update is permanent. When you use a change set, the result provides a diff of the updated stack and the original stack before you choose to execute the change.
Create or update a stack
Stack name
If you are updating an existing stack, choose the stack name.
cloudx2025-lambdastack
Activate Windows
Go to Settings to activate Windows.
Cancel
Done



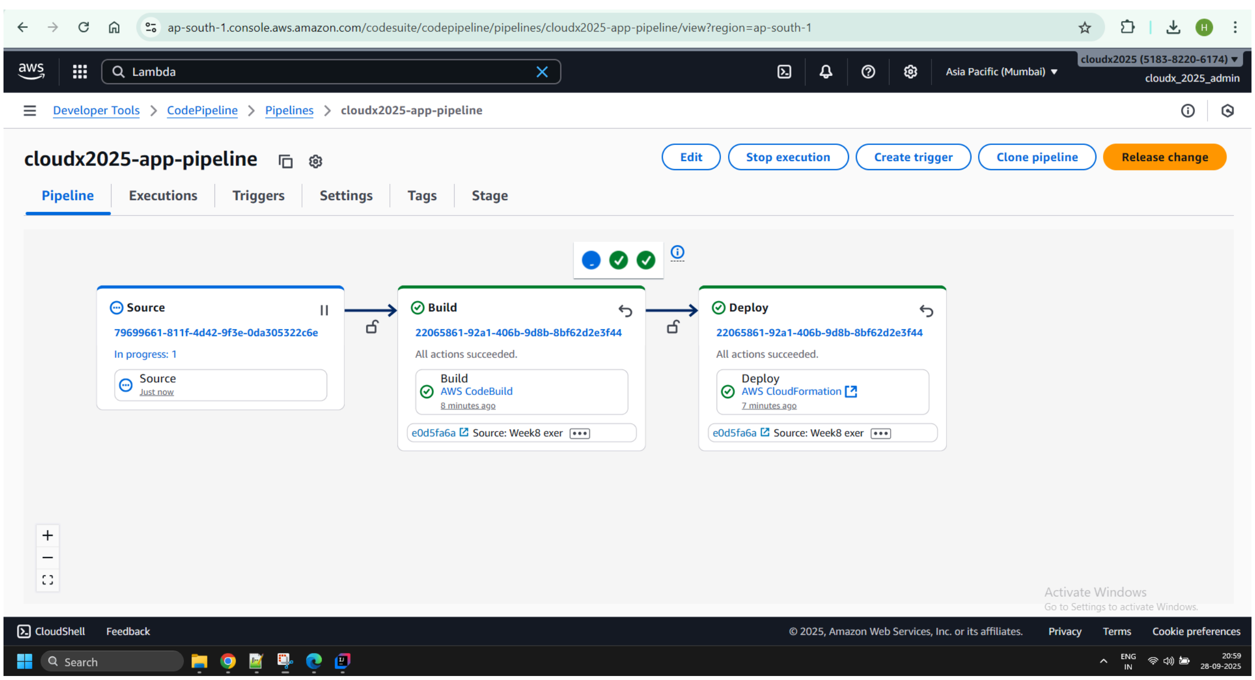
6. Delete the *<ProjectName>-UploadsNotificationFunction* Lambda function.

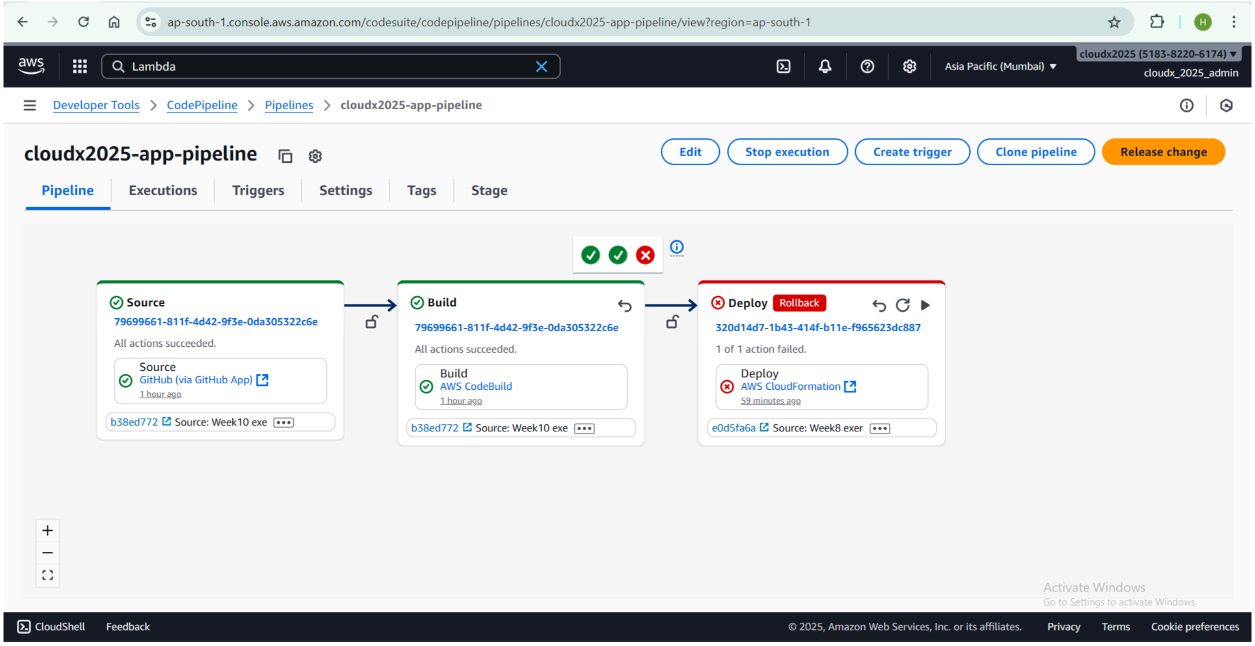


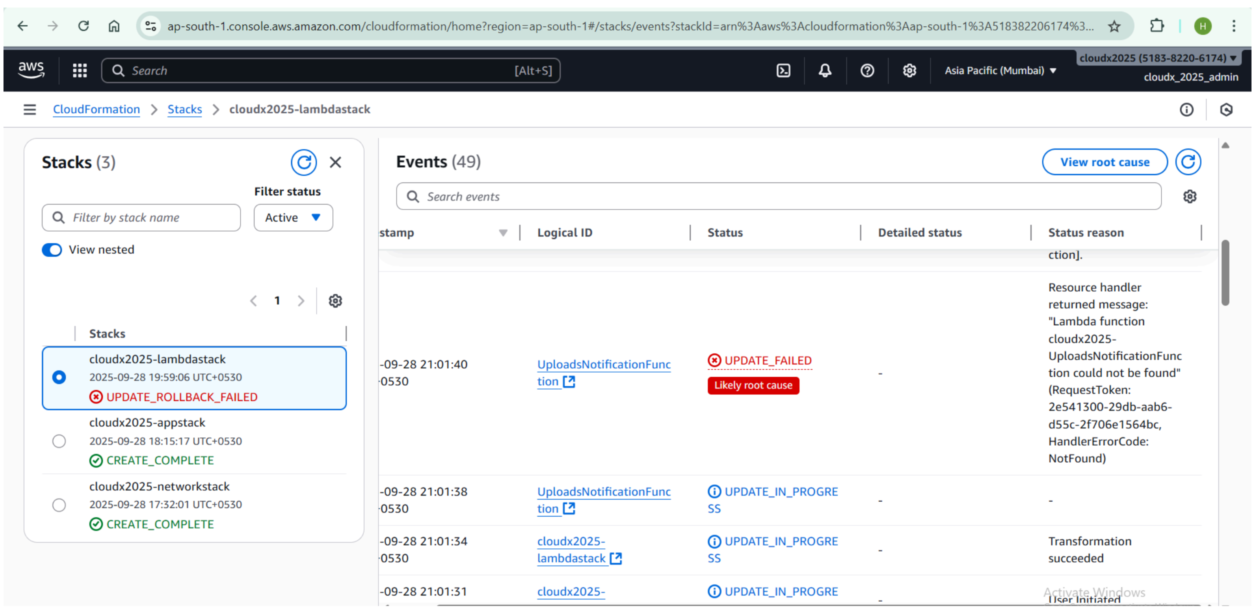
7. Make some changes to the Lambda function code and ensure the CodePipeline builds and deploys it.

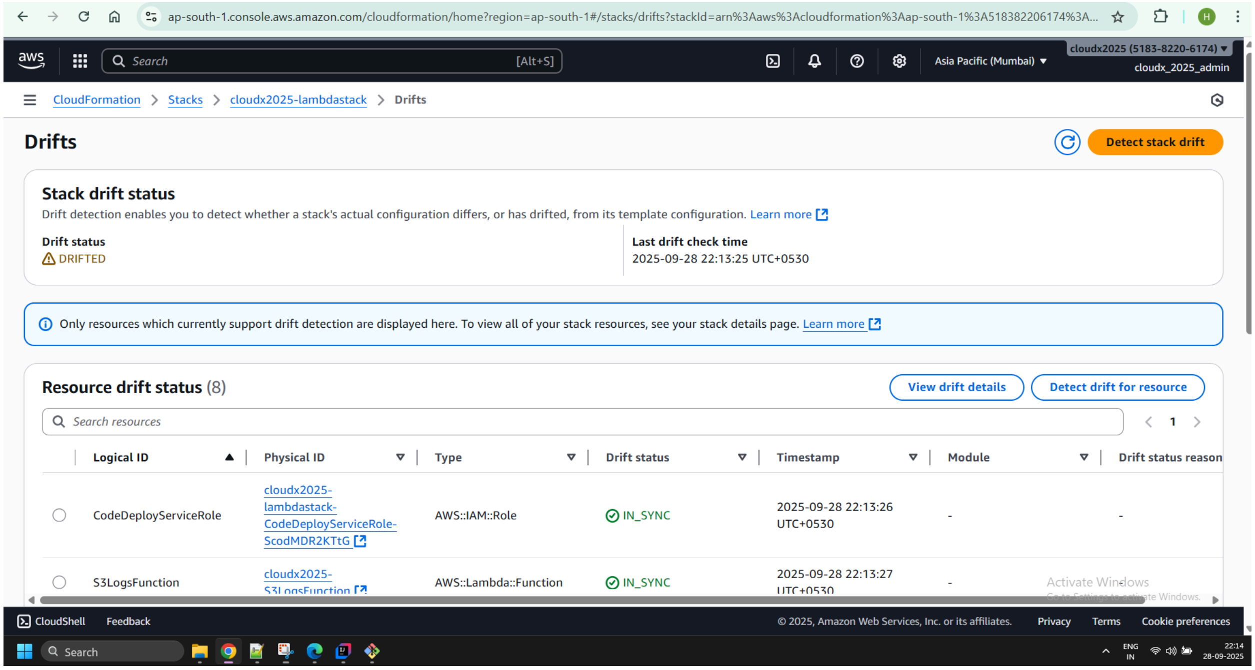


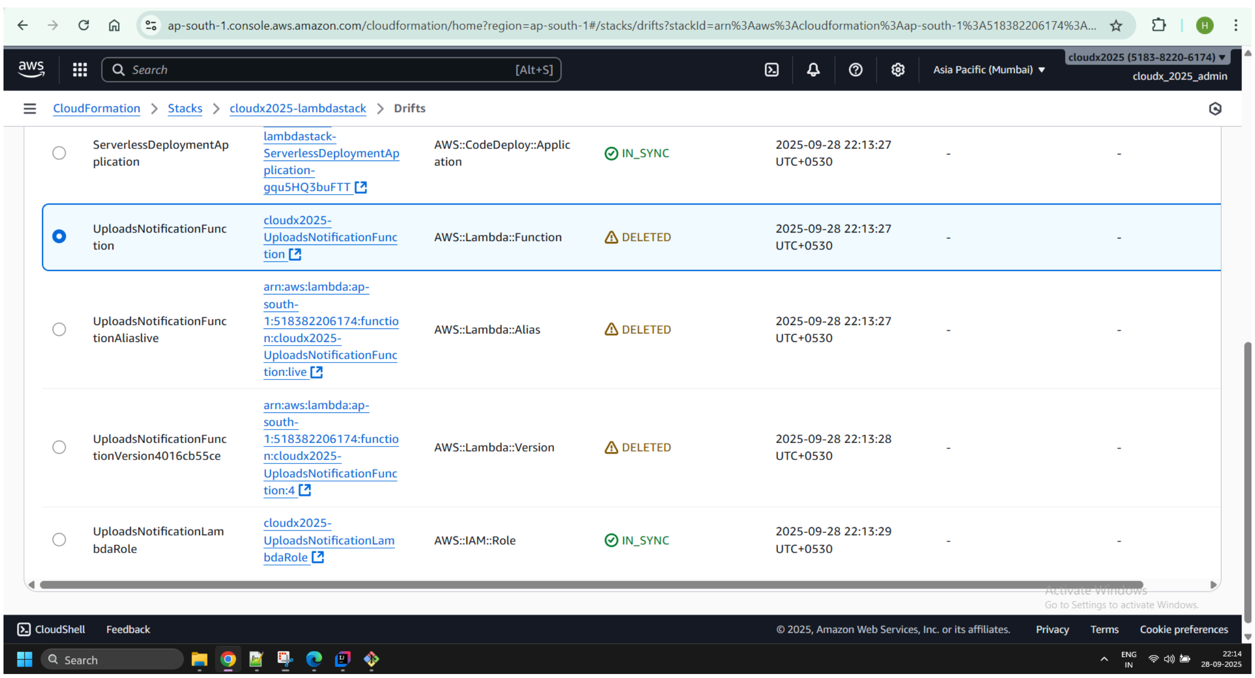
Pipeline gets triggered..



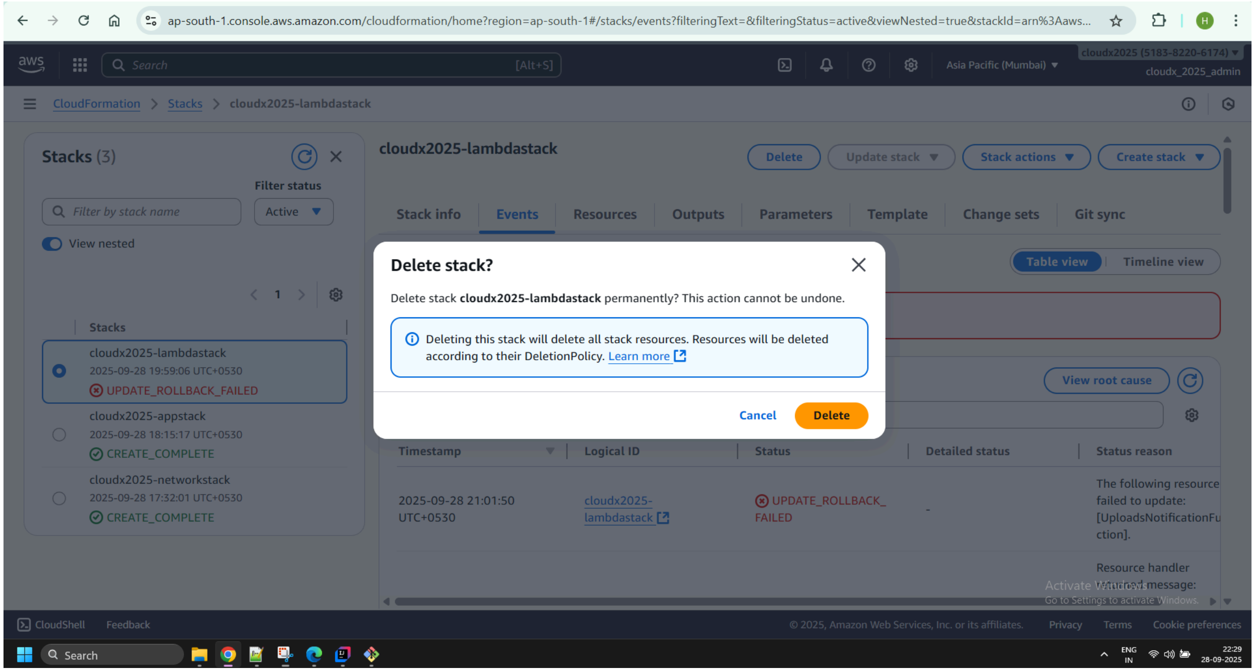

ap-south-1.console.aws.amazon.com/codesuite/codepipeline/pipelines/cloudx2025-app-pipeline/view?region=ap-south-1
Developer Tools > CodePipeline > Pipelines > cloudx2025-app-pipeline
cloudx2025-app-pipeline
Edit
Stop execution
Create trigger
Clone pipeline
Release change
Pipeline
Executions
Triggers
Settings
Tags
Stage
Source
79699661-811f-4d42-9f3e-0da305322c6e
All actions succeeded.
Source
GitHub (via GitHub App)
1 hour ago
b38ed772 Source: Week10 exe
Build
79699661-811f-4d42-9f3e-0da305322c6e
All actions succeeded.
Build
AWS CodeBuild
1 hour ago
b38ed772 Source: Week10 exe
Deploy
Rollback
320d14d7-1b43-414f-b11e-f965623dc887
1 of 1 action failed.
Deploy
AWS CloudFormation
59 minutes ago
e0d5fa6a Source: Week8 exer
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

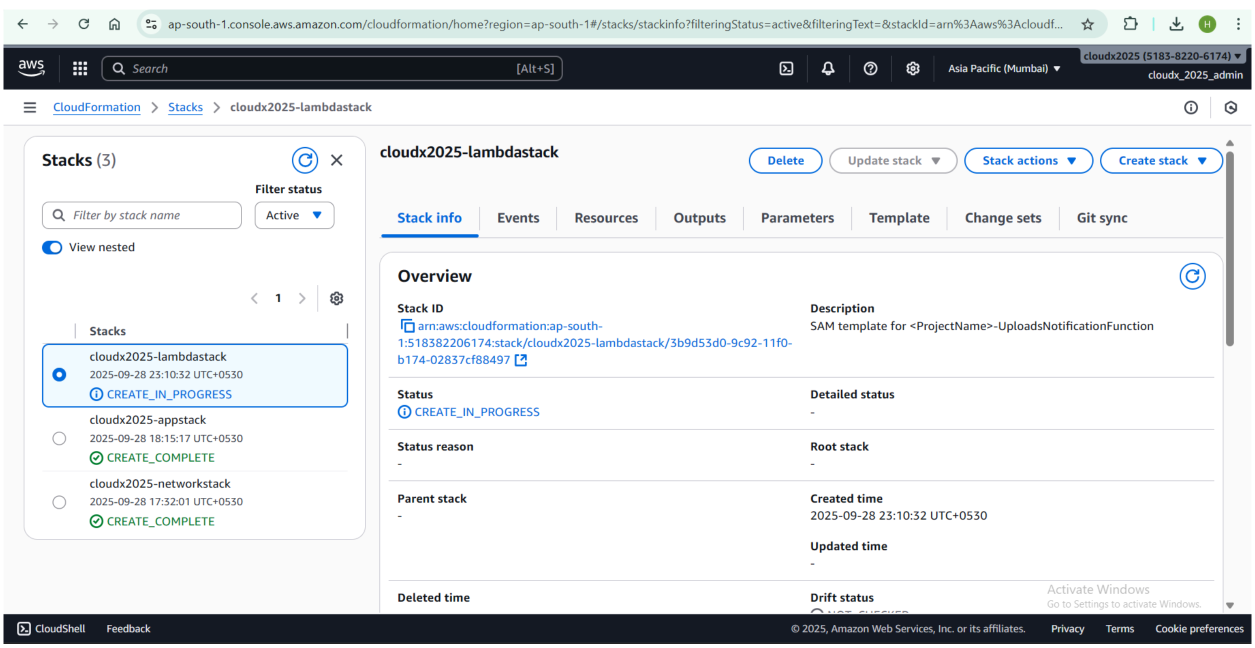

ap-south-1.console.aws.amazon.com/cloudformation/home?region=ap-south-1#/stacks/events?stackId=arn%3Aaws%3Acloudformation%3Aap-south-1%3A518382206174%3...
CloudFormation > Stacks > cloudx2025-lambdastack
Stacks (3)
Filter status
Active
View nested
Stacks
cloudx2025-lambdastack
2025-09-28 19:59:06 UTC+0530
UPDATE_ROLLBACK_FAILED
cloudx2025-appstack
2025-09-28 18:15:17 UTC+0530
CREATE_COMPLETE
cloudx2025-networkstack
2025-09-28 17:32:01 UTC+0530
CREATE_COMPLETE
Events (49)
View root cause
Logical ID
Status
Detailed status
Status reason
-09-28 21:01:40 0530
UploadsNotificationFunction
UPDATE_FAILED
Likely root cause
Resource handler returned message: "Lambda function cloudx2025-UploadsNotificationFunction could not be found" (RequestToken: 2e541300-29db-aab6-d55c-2f706e1564bc, HandlerErrorCode: NotFound)
-09-28 21:01:38 0530
UploadsNotificationFunction
UPDATE_IN_PROGRESS
-09-28 21:01:34 0530
cloudx2025-lambdastack
UPDATE_IN_PROGRESS
Transformation succeeded
-09-28 21:01:31
cloudx2025-
UPDATE_IN_PROGRE





Since the lambda resource was created initially using sam deploy's cloudformation stack so manually deleting it *outside* of the cloudformation stack has resulted in stack drift and hence pipeline issues. To fix this, we need to delete the lambda stack so that CodePipeline creates it for us.


ap-south-1.console.aws.amazon.com/cloudformation/home?region=ap-south-1#/stacks/events?filteringText=&filteringStatus=active&viewNested=true&stackId=arn%3Aaws...
cloudx2025 (5183-8220-6174)
cloudx_2025_admin
Asia Pacific (Mumbai)
CloudFormation > Stacks > cloudx2025-lambdastack
Stacks (3)
Filter status
Filter by stack name
Active
View nested
Stacks
cloudx2025-lambdastack
2025-09-28 19:59:06 UTC+0530
UPDATE_ROLLBACK_FAILED
cloudx2025-appstack
2025-09-28 18:15:17 UTC+0530
CREATE_COMPLETE
cloudx2025-networkstack
2025-09-28 17:32:01 UTC+0530
CREATE_COMPLETE
cloudx2025-lambdastack
Delete
Update stack
Stack actions
Create stack
Stack info
Events
Resources
Outputs
Parameters
Template
Change sets
Git sync
Table view
Timeline view
Delete stack?
Delete stack cloudx2025-lambdastack permanently? This action cannot be undone.
Deleting this stack will delete all stack resources. Resources will be deleted according to their DeletionPolicy. Learn more
Cancel
Delete
View root cause
Timestamp
Logical ID
Status
Detailed status
Status reason
2025-09-28 21:01:50 UTC+0530
cloudx2025-lambdastack
UPDATE_ROLLBACK_FAILED
The following resource failed to update: [UploadsNotificationFunction].
Resource handler
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


ap-south-1.console.aws.amazon.com/cloudformation/home?region=ap-south-1#/stacks/stackinfo?filteringStatus=active&filteringText=&stackId=arn%3Aaws%3Acloudf...
cloudx2025 (5183-8220-6174)
cloudx_2025_admin
Asia Pacific (Mumbai)
CloudFormation > Stacks > cloudx2025-lambdastack
Stacks (3)
Filter status
Filter by stack name
Active
View nested
Stacks
cloudx2025-lambdastack
2025-09-28 23:10:32 UTC+0530
CREATE_IN_PROGRESS
cloudx2025-appstack
2025-09-28 18:15:17 UTC+0530
CREATE_COMPLETE
cloudx2025-networkstack
2025-09-28 17:32:01 UTC+0530
CREATE_COMPLETE
cloudx2025-lambdastack
Delete
Update stack
Stack actions
Create stack
Stack info
Events
Resources
Outputs
Parameters
Template
Change sets
Git sync
Overview
Stack ID
arn:aws:cloudformation:ap-south-1:518382206174:stack/cloudx2025-lambdastack/3b9d53d0-9c92-11f0-b174-02837cf88497
Description
SAM template for <ProjectName>-UploadsNotificationFunction
Status
CREATE_IN_PROGRESS
Detailed status
-
Status reason
-
Root stack
-
Parent stack
-
Created time
2025-09-28 23:10:32 UTC+0530
Updated time
-
Deleted time
Drift status
CloudShell
Feedback
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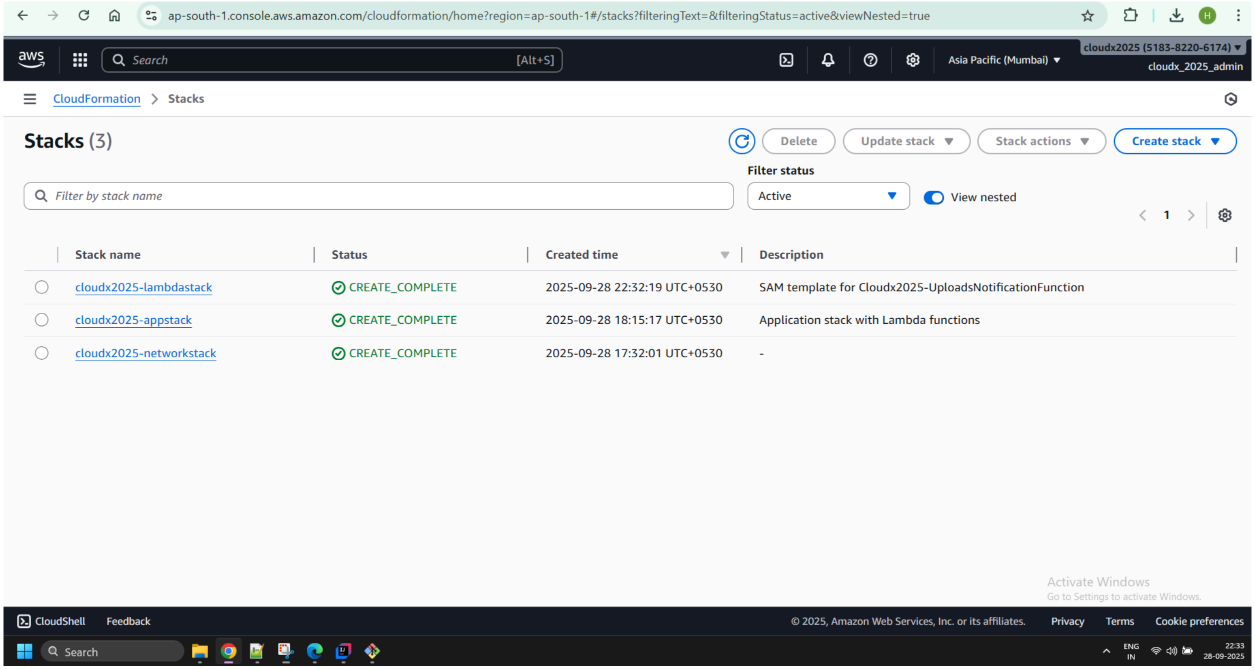
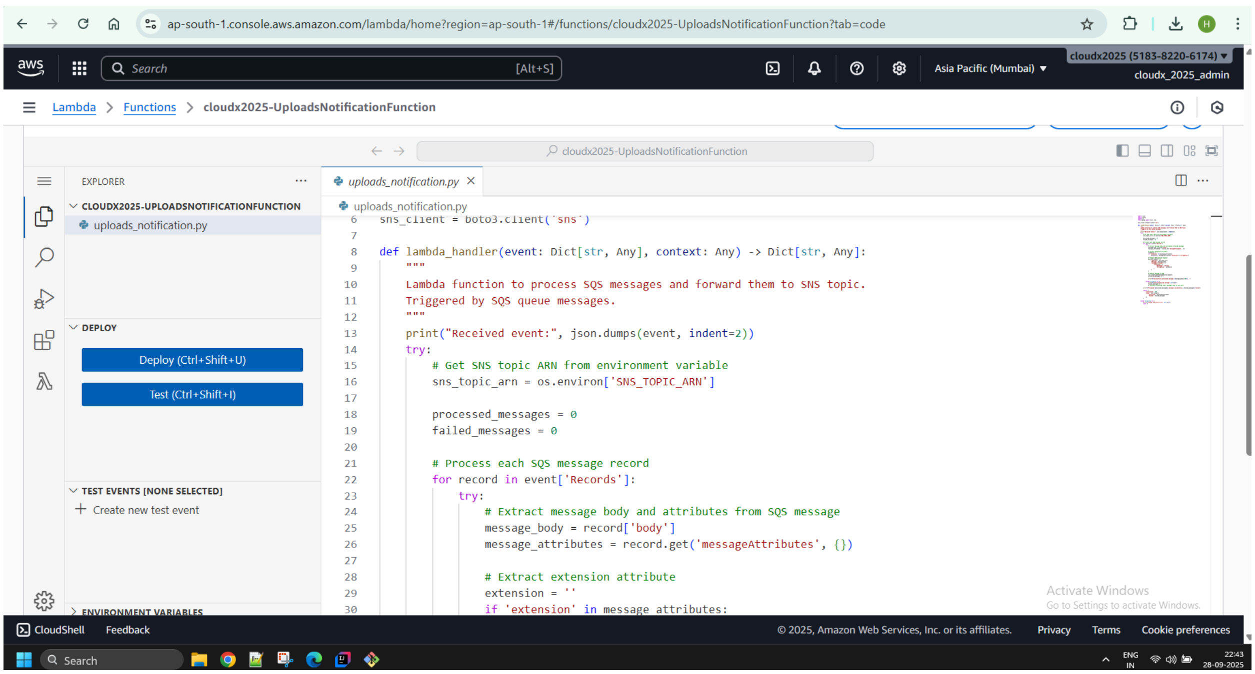
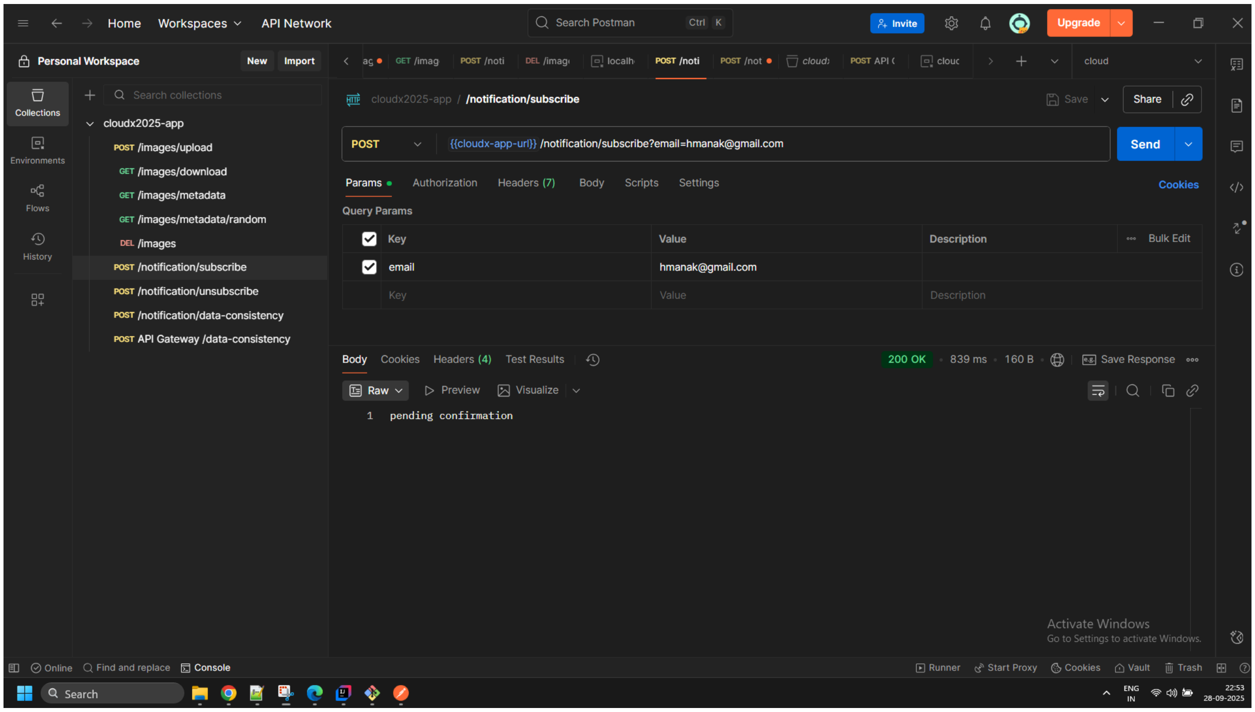
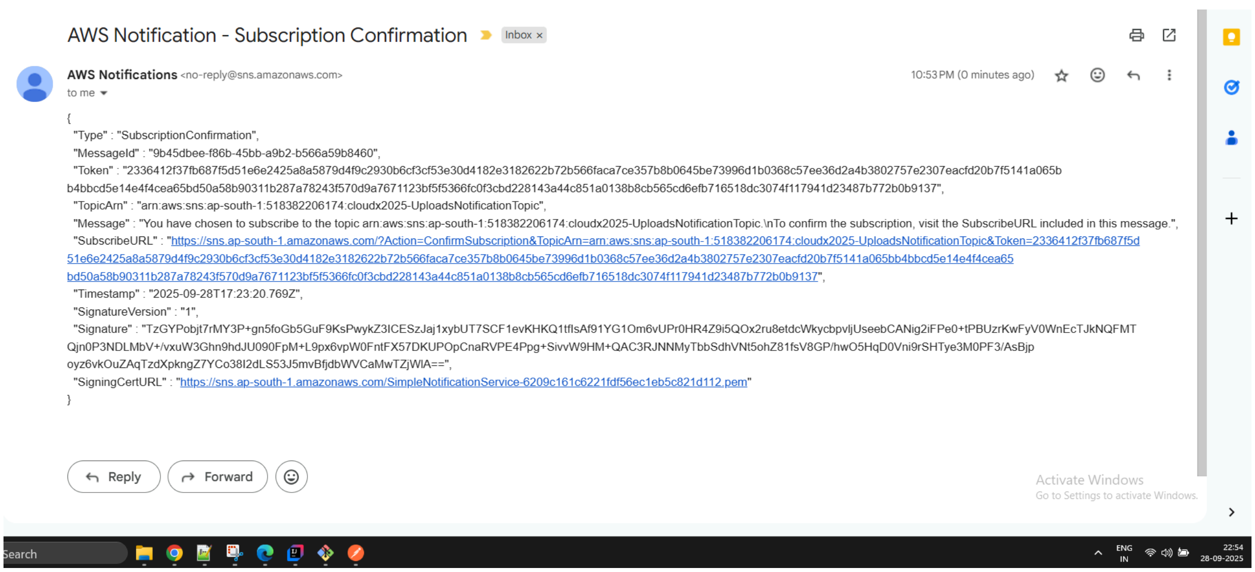
Privacy
Terms
Cookie preferences



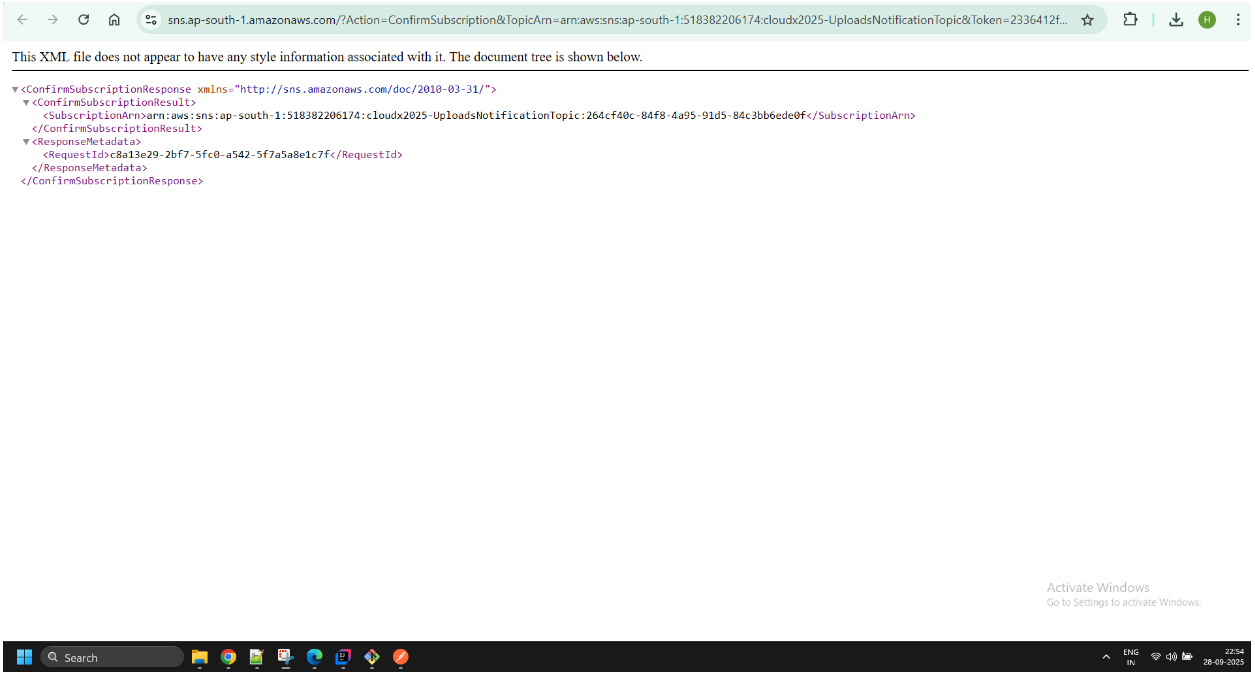


8. Upload a new image to the image S3 bucket using your web-application and ensure an email notification which mentions the image is sent.

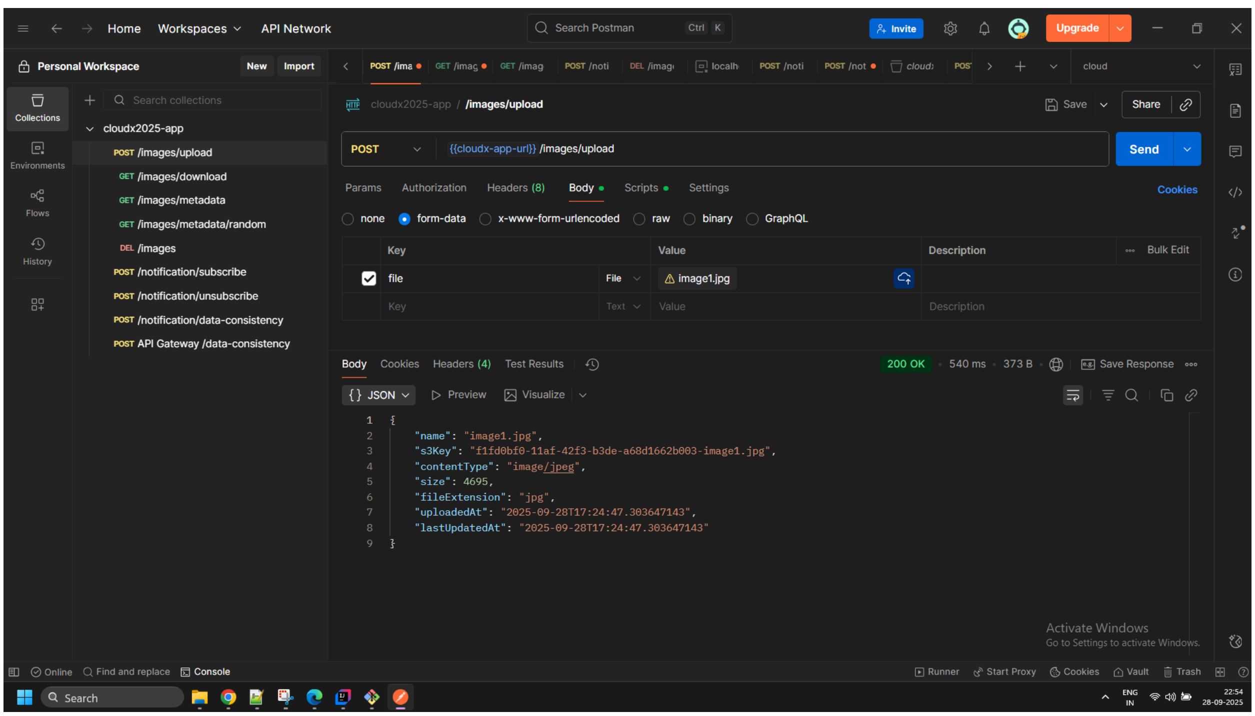




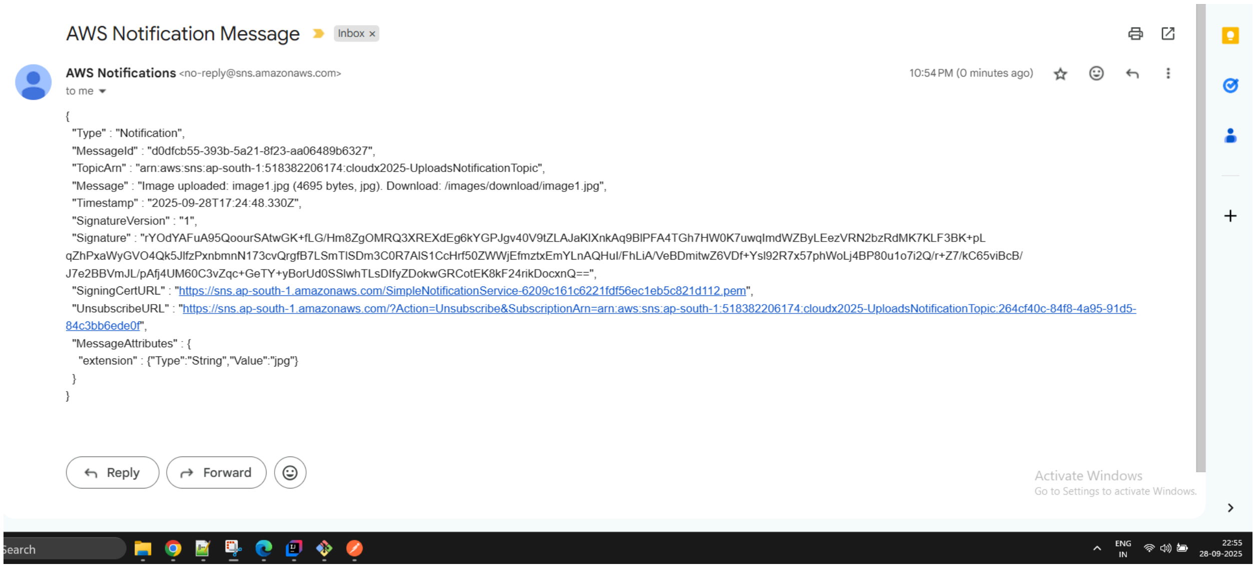

```
{
  "Type" : "SubscriptionConfirmation",
  "MessageId" : "9b45dbee-f86b-45bb-a9b2-b566a59b8460",
  "Token" : "2336412f37fb687f5d51e6e2425a8a5879d4f9c2930b6cf3cf53e30d4182e3182622b72b566faca7ce357b8b0645be73996d1b0368c57ee36d2a4b3802757e2307eacfd20b7f5141a065b
b4bbcd5e14e4f4cea65bd50a58b90311b287a78243f570d9a7671123bf5f5366fc0f3cbd228143a44c851a0138b8cb565cd6efb716518dc3074f117941d23487b772b0b9137",
  "TopicArn" : "arn:aws:sns:ap-south-1:518382206174:cloudx2025-UploadsNotificationTopic",
  "Message" : "You have chosen to subscribe to the topic arn:aws:sns:ap-south-1:518382206174:cloudx2025-UploadsNotificationTopic.\nTo confirm the subscription, visit the SubscribeURL included in this message.",
  "SubscribeURL" : "https://sns.ap-south-1.amazonaws.com/?Action=ConfirmSubscription&TopicArn=arn:aws:sns:ap-south-1:518382206174:cloudx2025-UploadsNotificationTopic&Token=2336412f37fb687f5d
51e6e2425a8a5879d4f9c2930b6cf3cf53e30d4182e3182622b72b566faca7ce357b8b0645be73996d1b0368c57ee36d2a4b3802757e2307eacfd20b7f5141a065bb4bbcd5e14e4f4cea65
bd50a58b90311b287a78243f570d9a7671123bf5f5366fc0f3cbd228143a44c851a0138b8cb565cd6efb716518dc3074f117941d23487b772b0b9137",
  "Timestamp" : "2025-09-28T17:23:20.769Z",
  "SignatureVersion" : "1",
  "Signature" : "TzGYPobjt7rMY3P+gn5foGb5GuF9KsPwykZ3ICESzJaj1xybUT7SCF1evKHKQ1tflsAf91YG1Om6vUPr0HR4Z9i5QOx2ru8etdcWkycbpvljUseebCANig2iFPe0+tPBUzrKwFyV0WnEcTJkNQFMT
Qjn0P3NDLMbV+/vxuW3Ghn9hdJU090FpM+L9px6vpW0FntFX57DKUPOpCnaRVPE4Ppg+SivvW9HM+QAC3RJNNMyTbbSdhVNt5ohZ81fsV8GP/hwO5HqD0Vni9rSHTye3M0PF3/AsBjp
oyz6vkOuZAqTzdXpkngZ7YCo38I2dLS53J5mvBfjdbWVCaMwTZjWIA==",
  "SigningCertURL" : "https://sns.ap-south-1.amazonaws.com/SimpleNotificationService-6209c161c6221fdf56ec1eb5c821d112.pem"
}
```



sns.ap-south-1.amazonaws.com/?Action=ConfirmSubscription&TopicArn=arn:aws:sns:ap-south-1:518382206174:cloudx2025-UploadsNotificationTopic&Token=2336412f...
This XML file does not appear to have any style information associated with it. The document tree is shown below.
<ConfirmSubscriptionResponse xmlns="http://sns.amazonaws.com/doc/2010-03-31/">
<ConfirmSubscriptionResult>
<SubscriptionArn>arn:aws:sns:ap-south-1:518382206174:cloudx2025-UploadsNotificationTopic:264cf40c-84f8-4a95-91d5-84c3bb6ede0f</SubscriptionArn>
</ConfirmSubscriptionResult>
<ResponseMetadata>
<RequestId>c8a13e29-2bf7-5fc0-a542-5f7a5a8e1c7f</RequestId>
</ResponseMetadata>
</ConfirmSubscriptionResponse>
Activate Windows
Go to Settings to activate Windows.
Search
22:54
28-09-2025



Home
Workspaces
API Network
Search Postman
Invite
Upgrade
Personal Workspace
New
Import
cloudx2025-app
POST /images/upload
GET /images/download
GET /images/metadata
GET /images/metadata/random
DEL /images
POST /notification/subscribe
POST /notification/unsubscribe
POST /notification/data-consistency
POST API Gateway /data-consistency
cloudx2025-app / /images/upload
POST
{{cloudx-app-url}} /images/upload
Send
Params
Authorization
Headers (8)
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
Key
Value
Description
Bulk Edit
file
File
image1.jpg
Body
Cookies
Headers (4)
Test Results
200 OK
540 ms
373 B
Save Response
JSON
Preview
Visualize
{
"name": "image1.jpg",
"s3Key": "f1fd0bf0-11af-42f3-b3de-a68d1662b003-image1.jpg",
"contentType": "image/jpeg",
"size": 4695,
"fileExtension": "jpg",
"uploadedAt": "2025-09-28T17:24:47.303647143",
"lastUpdatedAt": "2025-09-28T17:24:47.303647143"
}
Activate Windows
Go to Settings to activate Windows.
Online
Find and replace
Console
Runner
Start Proxy
Cookies
Vault
Trash
22:54
28-09-2025



Buildspec.yml file

```yaml
version: 0.2

phases:
 install:
   runtime-versions:
     java: corretto17
     python: 3.11
   commands:
     - pip install aws-sam-cli
     - echo Installing dependencies...
     - cd Week10/Task2
     - mvn clean install
 pre_build:
   commands:
     - echo Running tests...
     - mvn test
 build:
   commands:
     - echo Building jar...
     - mvn package
     - echo Packaging SAM template...
     - cd lambda
     - sam package --template-file sam-template.yml --s3-bucket $S3_BUCKET
--output-template-file sam-template-packaged.yml
artifacts:
 files:
   - Week10/Task2/lambda/sam-template-packaged.yml
   - Week10/Task2/target/*.jar
 discard-paths: yes
```

Codepipeline.yml file


```yaml
metadata:
 created: '2025-09-28T20:18:46.768000+05:30'
 pipelineArn:
arn:aws:codepipeline:ap-south-1:518382206174:cloudx2025-app-pipeline
 updated: '2025-09-28T20:41:30.327000+05:30'
pipeline:
 artifactStore:
   location: cloudx2025-harshitmanaktala-20250720-site
   type: S3
 name: cloudx2025-app-pipeline
 roleArn:
arn:aws:iam::518382206174:role/service-role/AWSCodePipelineServiceRole-ap-south
-1-cloudx2025-app-pipeline
 stages:
 - actions:
   - actionTypeId:
       category: Source
       owner: AWS
       provider: CodeStarSourceConnection
       version: '1'
     configuration:
       BranchName: main
       ConnectionArn:
arn:aws:codeconnections:ap-south-1:518382206174:connection/34b7c164-d4ef-4b15-8
fd3-6a02f200cca2
       DetectChanges: 'true'
       FullRepositoryId: peakit/aws-dev-associate
       OutputArtifactFormat: CODE_ZIP
     inputArtifacts: []
     name: Source
     namespace: SourceVariables
     outputArtifacts:
     - name: SourceArtifact
     region: ap-south-1
     runOrder: 1
   name: Source
 - actions:
   - actionTypeId:
       category: Build
       owner: AWS
       provider: CodeBuild
       version: '1'
     configuration:
       ProjectName: cloudx2025-app
     inputArtifacts:
     - name: SourceArtifact
     name: Build
     namespace: BuildVariables
```

```yaml
      outputArtifacts:
      - name: BuildArtifact
      region: ap-south-1
      runOrder: 1
    name: Build
  - actions:
    - actionTypeId:
        category: Deploy
        owner: AWS
        provider: CloudFormation
        version: '1'
      configuration:
        ActionMode: CREATE_UPDATE
        Capabilities: CAPABILITY_IAM,CAPABILITY_NAMED_IAM,CAPABILITY_AUTO_EXPAND
        ParameterOverrides: "{\n\"ProjectName\":
\"cloudx2025\",\n\"S3BucketName\"\
          :
\"cloudx2025-e572f1dd-4192-4f26-98c4-442d49f1d728\",\n\"SnsTopicArn\"\
          :
\"arn:aws:sns:ap-south-1:518382206174:cloudx2025-UploadsNotificationTopic\"\
          ,\n\"SqsQueueArn\":
\"arn:aws:sqs:ap-south-1:518382206174:cloudx2025-UploadsNotificationQueue\"\
          ,\n\"SqsQueueUrl\":
\"https://sqs.ap-south-1.amazonaws.com/518382206174/cloudx2025-UploadsNotificat
ionQueue\"\
          \n}"
        RoleArn: arn:aws:iam::518382206174:role/cloudformation-servicerole
        StackName: cloudx2025-lambdastack
        TemplatePath: BuildArtifact::sam-template-packaged.yml
      inputArtifacts:
      - name: BuildArtifact
      name: Deploy
      namespace: DeployVariables
      outputArtifacts: []
      region: ap-south-1
      runOrder: 1
    name: Deploy
  version: 5
```